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MSc in Data Science

Project Report

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From Prediction to Action: Personalized Explainable AI Profiles and Robustness Testing for At-Risk Students in Online Education

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Declaration

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Abstract

The challenge of student dropout in higher education is exacerbated by reliance on high-accuracy predictive models whose internal decision-making processes lack transparency. These systems often fail to provide the human-understandable justifications required for targeted institutional support. This dissertation addresses this critical need by developing a rigorous Explainable AI (XAI) framework, using the Open University Learning Analytics Dataset (OULAD), to ensure predictive systems are trustworthy and ethically governed.

The methodology employs a comparative analysis of a high-performance model (XGBoost) against a transparent linear benchmark (Logistic Regression) using the SHAP and LIME XAI techniques. The study makes three unique contributions: a temporal analysis revealing that predictive risk factors dynamically transition from demographic proxies to academic performance indicators over the semester; an algorithmic fairness audit showing that complex models can learn subtle, localized biases; and a robustness test.

The robustness evaluation yielded decisive results, confirming SHAP`s balance and proving LIME's volatility under minor data perturbation, which deems LIME unsuitable for high-stakes deployment. Finally, the framework translates version outputs into actionable student personas through clarification clustering. This study provides a vital basis for growing ethical, reliable, and policy-driven predictive structures in educational analytics.

Keywords

Keywords: Explainable AI (XAI), Student Dropout, Learning Analytics, SHAP, LIME, Algorithmic Bias, Robustness, OULAD, Time-Series.

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Chapter 1: Introduction

This chapter has research prerequisites by introducing the educational context, the research problem, and setting research goals. It begins with a discussion of the implications of development in predicting student performance and the role of explainable AI (XAI) in this field. Next, this chapter clarifies unique research questions that focus on the limitations of traditional models in academic settings. Finally, it presents research questions and goals related to my paper, and concludes with an overview of the report's structure.

1.1 Background

The growing availability of educational data, which is regularly referred to as gaining information from analytics, has opened up new possibilities for institutions to understand and rely on student behavior. In this period of large data, the ability to predict student outcomes, alongside academic achievement or the chances of dropping out, has turned out to be widely recognized in universities worldwide. Predictive models, powered through ML models getting to know algorithms, are becoming more and more widely used as a device to discover at-risk students, allowing educators to provide properly timed and centered interventions. The Open University Learning Analytics Dataset (OULAD) (Kuzilek et al., 2017), a publicly available, anonymized dataset from a huge distance-gaining information institution, provides a large aspect for studies. It carries a wealth of facts on student demographics, direction engagement, and academic overall performance, making it a great candidate for building and evaluating predictive models (Waheed et al., 2022). Various recent studies have efficiently leveraged this type of multidimensional time-series data to predict student overall performance in online learning environments (Wang et al., 2024).
However, regardless of the high predictive accuracy of many machine learning models (Lopez et al., 2024; Yang, 2021), their adoption in high-stakes fields like training has been slow. This reluctance stems from an extensive challenge, which includes the shortage of transparency in complex, "black-box" models (Ribeiro et al., 2016). For an educator, definitely understanding that a scholar is in risk is not enough; they want to understand why. Without a clean explanation, it's impossible to lay out effective interventions. This is where Explainable AI (XAI) emerges as a vital discipline. XAI seeks to bridge the distance between a model's predictive strength and human understanding, providing insights right into a version's decision-making process (Lundberg and Lee, 2017). My dissertation seeks to use a complete XAI framework for student dropout prediction, transforming a black-container version right into an obvious and trustworthy device for educators.

1.2 Problem Statement

The traditional machine learning models have proven advanced performance in predictive tasks, their internal activities are still largely opaque in a way that lack of transparency, often called "black boxes", which is an important obstacle to their practical application in education. An administrator cannot justify an intervention based on a model's prediction without their own justification. This blur has some important risks. First, the inability to trust. If there is no clear explanation, educators and students may not trust the predictions of the model, leading to low application rates. The predictions are considered arbitrary, based on faith in the system. Second, the inability to act. A prediction of "risk" is not an overview of exploitation. Educators should know what specific factors, such as low VLE commitments or poor school performance, stimulate predictions to design a significant intervention. Finally, there is a risk of deviation. Black box models can accidentally learn and maintain social prejudices in training data. Without the explanation frame, these prejudices could not be detected, resulting in unfair results for some student groups. The study directly solves this problem by developing a powerful method combining high-performance prediction models with a complete XAI framework. It goes beyond a unique static analysis to provide a dynamic vision and a lot of anticipation of a model. This approach is not only a technical exercise but also an important step to building ethical and reliable systems for educational institutions.

1.3 Research Questions

These are my dissertation research questions as follows:

1. How can a multi-version method to student dropout prediction, leveraging each interpretable and high-appearing models, be used to identify the important elements of student success within the OULAD dataset?
2. How do the explanations provided by SHAP and LIME differ in their ability to interpreting both a high-performing black-box model (XGBoost) and a transparent white-box model (Logistic Regression)?
3. To what extent do the predictive explanations from models for student dropout extend over the path of a semester, and what does this temporal evolution imply for the timing of educational interventions?
4. How do XAI techniques reveal potential algorithmic biases in a predictive model, and can these insights be used to audit for and understand disparities in predictions across sensitive demographic subgroups?

5. How stable and robust are the explanations provided by different XAI methods (SHAP and LIME) when a model's input data is slightly perturbed, and what does this imply about their trustworthiness for high-stakes decisions?
6. Can XAI explanations be used to section students into distinct, actionable personas, thereby translating complex model interpretations into a realistic framework for focused educational support?

1.4 Project Objectives

Based on the research questions, this dissertation aims to achieve the following objectives:

Objective 1: Developing and evaluating multi-model prediction frames. My goal is to focus on building, training and optimizing a range of models, including logistics, Xgboost, Random Forest and MLP. The goal is to compare their performance to predict the abandonment of students in Oulad data set and identify the most appropriate models to explain in the situation.

Objective 2: Perform an analysis of the comparison of the model's ability to interpret. This goal is for the purpose of comparing the forces and limits of the SHAP and LIMEin the ability to interpret both efficient performance that is XGBOOST and transparent white box model are defined as Logistics regression.

Objective 3: Implement dynamic frame to analyze time. This goal is aimed at implementing a method in a way that the importance of characteristics changes in a semester. The goal is to create and analyze weekly explanations of SHAP and LIME to understand the development over time of predictions and provide information at the time of educational intervention.

Objective 4: Complete audit for the algorithm for the features with demographic information. This goal focuses on the use of Ethics and responsible. The goal is to conduct equity audit by Xai techniques to identify and analyze all the differences in predictions through sensitive demographic groups, such as gender, age and disabilities.

Objective 5: Assess the reliability and strength of the interpretation. This goal aims to seriously assess the reliability of the explanations. I have made a disturbing analysis to check the stability of outings and LIME and make conclusions about their reliability on decisions in highlights.

Objective 6: Translate information into students who can exploit. This goal focuses on filling the gap between technical results and practical applications. The goal is to use clusters on explanatory values so that the student segment with separate characters, providing educators with a reality framework to provide targeted support.

That is all my goals will be seen in my project transparent.

1.4 Expected Outcomes

Based on the research problems and goals, this dissertation will produce some main results. The project will create a very effective prediction model for students, which can be based on the XGBOOST algorithm, with a defined ultrasound. This model will not only be accurate but also based on a full analysis, in which a quantitative compromise between the performance of the model and the ability to interpret.

The result of this analysis will prove that a black box model can be explained effectively; this is an important step for its practical application in areas with high problems such as education. The project will also create a series of visual images of the time series showing the dynamic development of the importance of the characteristics in a semester. These visuals will highlight a clear and systematic change in demographic characteristics, which is very important from the beginning, to academic elements, becoming dominant in the course.

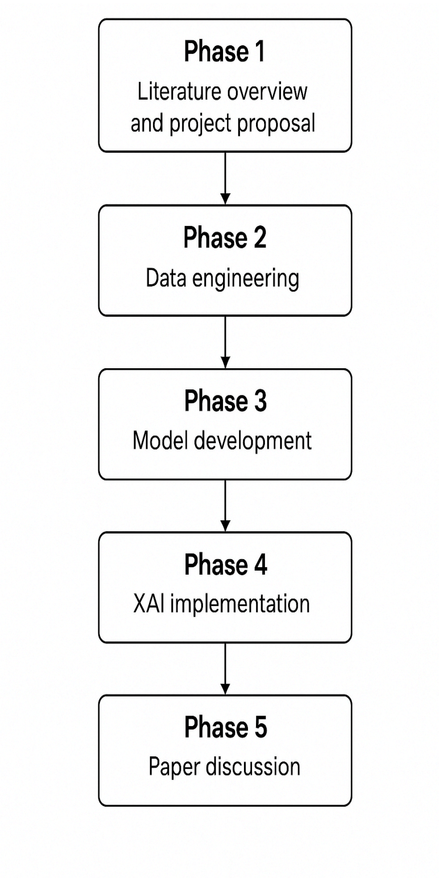
In addition, an equity audit will identify and record all potential algorithm deviations in models, providing a basis for determining who is responsible for education. Finally, the project will create a powerful analysis that offers an important assessment of the reliability of SHAP and LIME, showing that not all XAI methods are reliable in the presence of data disturbance. The clustering of model explanations will result in a set of distinct student personas, which can serve as a practical tool for educators to design targeted interventions. These personas will translate complex technical findings into a human-centered framework for educational support.

1.5 Purpose and Key Beneficiaries

The core purpose of this project is to address the gap between the predictive power of machine learning and the need for human-understandable explanations in high-stakes domains like education. Although current models can accurately predict student results, their nature hinders the application and use of their effectiveness. This dissertation provides a complete and transparent method, not only reaching high prediction accuracy, but also provides a clear understanding and can exploit the stimulating factors of these predictions. This brought the field beyond a pure technical exercise to a more moral application and focused on the man in data science.

The Key beneficiaries of this study are many. Educators and administrators will earn a transparent and reliable tool to identify students at risk and a framework to design targeted, timely and effective interventions. Research will also benefit students by leading to more fair and personal education support, eventually contributing to reducing the rate of giving up and improving learning results. Finally, the academic community will benefit from a new method with many aspects to apply to educational data, providing an important basis for future research in this field.

1.6 Methods and Work Plan



The project workflow is organized in five different phases, as shown in Figure 1 of the flow diagram. Phase 1 begins with the first study, a comprehensive literature overview and development of project proposals. This ensures that the research is based on the relevant academic and practical context. Phase 2 focuses on data engineering, including data preparation of OULAD data records through preprocessing, merging, and cleaning, feature engineering, and data preparation of the final data set. These steps are important to ensure that the data is reliable, consistent, and suitable for subsequent modeling tasks. Phase 3 is dedicated to model development, with four predictive models being trained, optimized, and evaluated. From these, the two best performance models have been selected for a deeper analysis of phase 4, focusing on Xai implementations. Using SHAP and LIME, the model is interviewed, provides temporary explanations, checks for fairness, conducts robustness tests, and identifies wise student personas through clustering. Finally, in phase 5, the paper is critically discussed, focusing on the results, limitations, and conclusions in which the results are synthesized, and the paper is prepared for submission. This step-by-step approach provides a clear, logical, and structured way to achieve project goals.

Figure 1:Describing my workflow for study.

1.7 Dissertation Structure

This dissertation is held in six chapters. Chapter 2 provides a complete review of relevant academic materials, including students' success predictions, Xai, and OULAD data sets. Chapter 3 Details of research method, data preprocessing, and model selection to make an advanced XAI framework. Chapter 4 Presentation of objective results of model performance and XAI analysis, supported by tables and data. Chapter 5: Discuss the meaning of these results, compare them with existing literature, and discover their practical meaning. Finally, Chapter 6 ends the dissertation by summarizing achievements, describing the main contributions, providing important self-assessments, and proposing paths for future research.

1.8 Generative AI Use

This research strictly adhered to the University`s policy regarding the use of Generative Artificial Intelligence (GenAI) tools, specifically [e.g., Google Gemini, Chat GPT] and [Paint for creating flow charts, GitHub Copilot]. GenAI becomes employed completely as a productivity resource for non-analytical tasks, making sure that each one center's intellectual contributions and novel analytical findings continue to be the only work of the writer. Assistance is broadly refined in the Conclusion, clarifying complicated instructional flow.

In the methodological and technical phases, GenAI becomes limited to producing Python functions and standard application code for routine processes. This includes putting in fundamental version pipelines, loops for k-fold cross-validation, and imposing foundational information preprocessing steps like one-hot encoding. This use aimed to improve performance in imposing trendy practices. Crucially, the tools have been restrained from influencing the core intellectual property of the venture, such as the layout of the perturbation analysis or the translation of the XAI results.

All important analytical work is developed and executed independently by the author. This consists of the implementation of the SHAP TreeExplainer, the calculation of temporal feature importance, the bespoke perturbation evaluation script, and the whole K-Means clustering set of rules implemented to the SHAP values. By strictly isolating those novel analytical additives from recurring generation, the venture guarantees that the integrity and originality of the study's findings are completely preserved and auditable.

In short, this chapter has introduced the educational context, clarifying the research issues and describing the main goals and contributions of my project. i have emphasized the importance of predicting students' results by using automatic learning, while highlighting the important need for transparency through Explainable AI. By presenting research problems, goals, expected results, and work processes, this chapter created a clear basis for the research. The next chapter is based on this basic task and begins with a complete review of the related tasks in Chapter 2.

Chapter 2: Critical Context & Literature Review

This chapter offers an important assessment of existing academic literature on students' success predictions and Explainable Artificial Intelligence (XAI), establishing the theoretical context of this thesis. The objective is to sum up the current knowledge, to identify the main research gaps, and justify the unique methods of this study. The examination of the structure of the first context of the issue of dropout students in higher education, followed by a detailed exploration of prediction and explanation techniques.

2.1 Student Success Prediction

Predicting students' learning results, such as performance, retention, and dropout rates, has become the central target of educational data mining (EDM) and learning analysis (LA) in recent years. The use of large data sets such as the Open University Analysis (OULAD) allows researchers to apply a range of machine learning techniques (ML) to this problem. Initial research in this field, as shown by Fernandes et al. (2020), mainly focuses on building predictive models with different types of models and their accuracy.

This project, using a set of data with features similar to Oulad, shows the feasibility of using ML to provide student results. Similarly, Waheed et al. (2022) provide a valuable overview of the various models, from traditional methods like SVM and K-Nearest Neighbour to more advanced deep learning techniques that have been applied to the OULAD dataset.

However, a significant limitation of a large part of this first study is its only predictive performance. Like Wang et al. (2024), very accurate models, especially complex learning architectures like the LSTM network, often act like "black boxes". Although these models are effective in identifying models in the time series related to students' participation, their opaque nature makes it confusing for educators to understand basic reasons for predictions.

This transparency creates an important research gap because it limits the ability of educational institutions to design effective and targeted interventions. The literature, therefore, points to a clear need for a methodological shift from merely predicting outcomes to explaining them, a need that is addressed by the field of Explainable AI (XAI) brings novelty to the study.

2.2 Introduction to Explainable AI (XAI)

The complex and high-performing “black-box” models have led to a growing demand for transparency in AI. Explainable AI (XAI) is a multidisciplinary field of study that seeks to address this demand by providing methods and techniques to make the decisions of AI systems understandable to humans. The significance of XAI is particularly acute in high-stakes domains such as education, where a model’s prediction of student failure could have life-altering consequences (Kumar, Singh and Sharma, 2025). In these contexts, merely knowing that a student is “at-risk” is insufficient; a proper intervention requires understanding the specific factors contributing to that risk.

XAI techniques are often categorized by the type of explanation they provide: global or local. Global explanations aim to provide a comprehensive, model-wide understanding of how the model works. This typically involves identifying the most important features across the entire dataset. Local explanations, on the other hand, focus on a single prediction, explaining why the model made a specific decision for an individual instance. This distinction is critical in education: global explanations can inform institutional policy (e.g., a curriculum change), while local explanations can inform a specific teacher-student intervention (e.g., a conversation with a student about their low engagement).

2.3 XAI Techniques

The present research leverages two of the most influential and widely adopted XAI techniques: SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations).

2.3.1 SHAP (SHapley Additive exPlanations)

Introduced by Lundberg and Lee (2017), SHAP is a unified framework for interpreting model predictions. It is based on the concept of Shapley values from cooperative game theory, which provides a theoretically sound method for distributing a model’s total output (the “payout”) among its input features (the “players”). The SHAP value for a feature represents its average marginal contribution to the prediction across all possible combinations of features. SHAP is distinguished by its strong theoretical guarantees, with explanations that are both consistent and accurate. A key strength is its unified approach, which enables the provision of both local explanations for individual predictions and global insights into the overall model behavior.

Formally, the Shapley value for a feature i is defined as:

$$\phi_i(f,x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!} \left[f_x(S \cup \{i\}) - f_x(S) \right]$$

where:

- F is the set of all features,
- S is a subset of features not containing i,
- $f_x(S)$ is the model prediction given only the features in S.

This formula ensures that contributions are distributed fairly across all features by considering all possible subsets. SHAP is distinguished by its strong theoretical guarantees, with explanations that are both consistent and accurate.

While SHAP is model-agnostic, for tree-based ensemble models like XGBoost, optimized implementations exist. These specialized versions significantly reduce the computational burden, making the exact calculation of SHAP values more feasible for large-scale projects. However, despite these optimizations, SHAP's exact calculation remains computationally intensive and can be slow for complex models, representing a practical limitation that must be carefully considered during project design (Zarei and Pahl, 2025).

2.3.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME, as introduced by Ribeiro, Singh and Guestrin (2016), provides local interpretability for any machine learning model. The core idea is to approximate the behavior of a complex, opaque model around a specific instance with a simpler, more interpretable model, such as a linear regression. By generating a new dataset of perturbed instances and their corresponding predictions, LIME trains this surrogate model to explain the black-box model's behavior in that local region.

Formally, LIME learns an interpretable model g that approximates the original model f locally around the instance x by solving:

$$\xi(x) = \arg\min_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

where:

- G is the class of interpretable models (e.g., linear models, decision trees),
- $\mathcal{L}(f, g, \pi_x)$ measures how close the explanation g is to the original model f, weighted by the locality measure π_x,
- π_x defines the proximity of the perturbed samples to x,
- $\Omega(g)$ is a complexity penalty to ensure g remains simple and interpretable.

A primary advantage of LIME is its model-agnostic nature, making it highly flexible and capable of explaining any model's predictions. The explanations provided by LIME are intuitive and easy to understand, as they are based on a simple local model. Additionally, LIME is often faster to compute than SHAP for a single explanation, making it a viable option for quick, on-the-fly interpretations (Liu, Zhou and Liu, 2025). However, a major limitation of LIME is the instability of its explanations. The results can be sensitive to the random sampling of the perturbed data points, meaning that running the method twice on the same instance can yield different results. This lack of stability can undermine trust in the explanation, a crucial consideration in high-stakes domains like education (Prümper, von Rymon Lipinski and Roth, 2023).

2.4 The OULAD Dataset

The **Open University Learning Analytics Dataset (OULAD)** is a publicly available, anonymous dataset compiled by Kuzilek, Hlosta, and Zdrahal (2017) to facilitate data discovery and learning analysis. It is an ideal resource for this dissertation due to its comprehensive nature and direct relevance to the issue of student dropout in online learning. The dataset is derived from 32,593 students across 22 presentations of seven distinct modules.

The data is structured across seven interconnected CSV files, providing a holistic view of student success factors. Key features include Demographic Data (e.g., gender, age band), VLE Interaction Data (capturing daily clicks on materials), and Assessment Data (scores and submission dates). This combination of behavioral, academic, and sensitive demographic data is crucial for two main reasons: firstly, it allows for the construction of robust predictive models; and secondly, it enables a detailed audit of fairness and bias, which is an important ethical consideration in predictive analysis.

2.5 Research Gaps

This dissertation is positioned to make several unique contributions by addressing key gaps in the existing literature:

- **Lack of Integrated XAI Analysis:** Much of the existing research on the OULAD dataset focuses either on predictive performance (e.g., Fernandes et al., 2020) or on XAI methods in isolation. My research combines a comparative analysis of SHAP and LIME to provide a more holistic view of explainability, bridging the gap between model accuracy and interpretability.

- **Absence of Temporal Analysis:** The literature largely treats student data as static, yet student behavior is dynamic and evolves over time. My dissertation introduces a novel temporal explainability analysis, directly addressing the limitations of models that struggle with interpretability over time (Wang et al., 2024). This provides new insights into how the factors influencing student success change throughout a semester.

- **Unproven Robustness of Explanations:** The stability of LIME’s explanations is a known weakness (Ribeiro, Singh and Guestrin, 2016; Prümper, von Rymon Lipinski and Roth, 2023), and the trustworthiness of explanations in general is a growing concern (Liu, Zhou and Liu, 2025). My research fills this gap by conducting a perturbation analysis to empirically test and validate the robustness of both SHAP and LIME explanations, providing a new methodological benchmark for the field.

- **Limited Translation of Insights to Action:** While papers like Fernandes et al. (2020) focus on predicting at-risk students, they often fail to translate these insights into actionable strategies for educators. The study addresses this by developing a methodology for clustering explanations to create actionable student personas (Cooper, 2022). This translates complex model outputs into practical, policy-driven solutions, a critical step that previous research has not fully realized.

- **Limited Fairness Auditing:** The literature highlights the importance of fairness auditing (Kumar, Singh and Sharma, 2025), but there is a lack of rigorous, multi-methodological approaches applied to datasets like the OULAD. This dissertation addresses this by performing a comprehensive fairness and bias audit using both SHAP and LIME to analyze disparate impacts across demographic subgroups, providing a more transparent and rigorous framework for ethical AI in education.

Summary of Chapter 2

This chapter successfully establishes the theoretical and empirical context for the research. It synthesizes existing knowledge on student dropout prediction and XAI, clearly identifying the limitations of prior work. By positioning this dissertation to address a crucial need for a dynamic, robust, and ethical XAI framework, the chapter provides a strong justification for the unique methodological approach outlined in the next section. The insights from this literature review provide a solid foundation for the research questions and objectives that will drive the rest of the project.

Chapter 3: Method

This chapter describes systematic research designs and describes a systematic framework from data preparation to multi-model-XAI implementation. The Oulad data records were cleaned, converted, and formatted for machine learning (Kuzilek et al., 2017), followed by strategic selection of four predictive models. Each model was subject to baseline and hyperparameter tuning. This chapter highlights explanatory AI and shows how logistics regression and XGBoost are used not only for forecasting but also for interpretable analyses, there by determining the basis for the findings presented in the next chapter.

3.1 Data Preprocessing

This section describes the important and basic processes of data preparation and provides careful steps to convert raw, multi-file, open university learning analytical data records (OULAD) into a uniform, clean, model-enabled format. This phase is of paramount importance to the paper, as data quality, structure, and integrity directly affect the performance and reliability of predictive models, as well as the findings that follow. OULAD is a published, anonymized resource compiled to facilitate research on educational data dates (Kuzilek et al., 2017). This includes information on 32,593 students in 22 presentations in seven different modules. The raw data consists of seven interconnected CSV files. This requires a robust integrated strategy to create an overall and accurate representation of student behavior and performance for model training.

3.1.1 Data Integration and Raw File Configuration

In the data integration phase, we covered 7 different CSV files, merging using common names (such as code_module, code_presentation, and id_student) to create comprehensive observations at the student level. A detailed understanding of the contributions of each file was essential.

1. StudentIninfo.csv: Provide fundamentally demographic data (gender, region, highest education, age band, disability status). It was important to include final results for all students that serve as the basis for the target variables.
2. StudentRegistration.CSV: Documented registration and unregistered data that occur in time to understand the duration of student commitment.

3. Assessments.csv and StudentAssessment.csv: These files are combined to quantify academic achievement. They detailed the assessment structure (type, weight, cutoff data), recorded the scores students achieved, and established a direct link between the work submitted and the final results.
4. Course.CSV: acts as a reference for the modulus and presentation codes and provides the necessary context information.
5. VLE.CSV and StudentVLE.CSV: These files were centrally important for capturing student behavior. Detailed study materials and activity types for vLe.csv.
6. StudentVLE.CSV provided a detailed protocol for student interactions and recorded clicks on specific materials on a particular day.

These interaction data were fundamental to the creation of behavioral features used in both predictive modeling and XAI clustering.

3.1.2 Missing Data Handling and Imputation Strategy

A systematic, step-by-step approach was implemented to process missing data and ensure data integrity. The initial step was to see missing values for my dataset. A concentration of values on properties was done, such as IMD_BAND (indexes of multiple drawers) and various VLE interaction columns. The strategy began with exclusion to two levels. By excluding properties, a high percentage of shortages (over 30%) of entries were completely removed to prevent the introduction of important assignment-based distortions from highly incomplete features. Exclude instances. This removed the row with the last_Result (target variable) missing. The remaining properties used double-track strategy assignment: numerical assignment sets are in the lack of values (particularly within the action and evaluation columns) (particularly within the action and evaluation columns). Finally, as category assignment replaced missing values for the remaining categories with "unknown" categories, the machine learning model was able to explicitly learn the predictive relationships of errors instead of assumptions about the most common categories

3.1.3 Feature Engineering: Target Creation and Aggregation

Functional engineering was an important and necessary step in transforming raw time series and categorical data into functional way as to optimized for my machine learning models

- **Creating the Binary Target Variable (y):** The study's objective is to predict student dropout risk. The final_result column was transformed into a binary variable.

Students with a final_result of 'Withdrawn' or 'Fail' were labeled 1 (Dropout). This grouping strategically combines voluntary and academic failure into a single target class, representing all students who did not successfully complete the course with a passing grade. Students labeled 'Pass' or 'Distinction' became 0 (Non-Dropout).

- **Aggregating VLE Activity (Behavioral Features)**: The granular, day-by-day VLE interaction data (studentVle.csv) was aggregated using the groupby() method in the pandas library. This converted the time-series activity logs into key student-level, static features that act as proxies for engagement and behavior (Wang et al., 2024). The engineered features include total_clicks (cumulative engagement), avg_clicks (average daily intensity), num_interactions (frequency of access), and unique_vle_materials (breadth of content usage). These synthesized features are particularly useful as they capture the temporal aspects of learning and will be central to the subsequent XAI analysis.

3.1.4 Final Dataset Formatting and Preparation

The final step in the data preprocessing process has focused on ensuring data sets and structural compatibility with the machine algorithms selected. To achieve this, the One-Hot Encoding was applied to all nominal classification variables (for example: gender, region, education level). This is essential to convert them into digital formats without the order of different types. After that, the selection of features was done in where the unpredictable columns were deleted; More specifically, the ID_Stitude column has been deleted to prevent over-adjusting models with arbitrary locks, and the original column has been deleted to avoid data leakage. Finally, for monitoring learning, the data kit is officially divided into two components: Characteristic Matrix (x), containing all independent variables, and target vectors (Y), containing binary labels, setting the final prerequisite for training and model evaluation stages

3.3 Model Selection

The model selection process for this research is designed to balance the predictable performance with the ability to interpret while ensuring strong evaluation through various automatic learning methods (Lopez et al., 2024). The work process for the development of the model is illustrated in Figure 2.

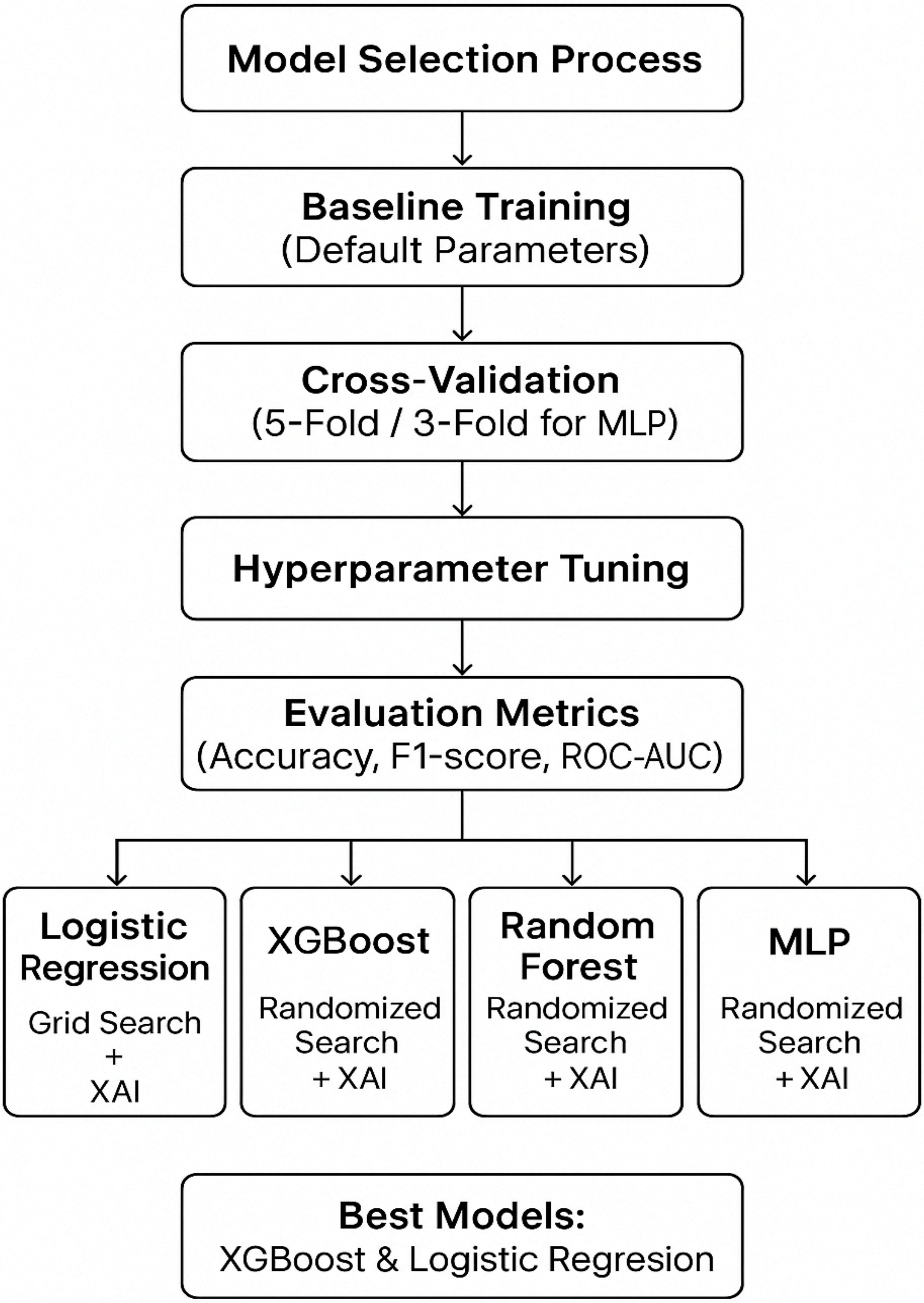


Figure 2 The flowchart summarizes this approach and the model-specific strategies

Each model was systematically developed using best practices. Basic training (default settings) established the initial performance reference. All models then underwent cross-validation (5-fold stratified, or 3-fold for MLP) to ensure robust performance estimates and mitigate class imbalance. Optimization was achieved via Hyperparameter tuning (Randomized Search or Grid Search), maximizing the F1-score while controlling model complexity. Evaluation Metrics included Accuracy, F1-Score, and ROC-AUC for a holistic assessment. Finally, Model-Specific Optimization & XAI involved tailored tuning and subsequent explanation using SHAP and LIME, ensuring both predictive power and transparency.

3.3.1 XGBoost

XGBOOST (extremely high slope stimulating) has been chosen as the first prediction model for this thesis, taking advantage of its proven capabilities to grasp complex and nonlinear relationships and offer strong prediction efficiency in student prediction (Yang, 2021). As an overall method based on effective and effective trees, widely applied in the industry and research, its compatibility with explanatory frames such as SHAP and LIME has further enhanced its role, allowing assessments based on accuracy and interpretation (Nagy and Molontay, 2023).

A basic XGBClassifier was originally formed by default parameters and was assessed through a cross-stratified validation at 5-fold. This process has established an efficiency reference by using accuracy, F1, and ROC-AUC points, with the stratification to ensure that the imbalance of the layer in the abandoned data set has been preserved to provide reliable estimates.

3.3.1.1 Hyperparameter Tuning

The optimization is made using random search with cross-validation for 5-fold stratification, an effective option that has been used effectively in exploring a superhomogeneous space compared to research on the grid. The adjustment focuses on the main parameters (n_estimators, max_depth, learning_rate, sub-sub, and college_bytree), directly controlling the complexity of the model and excessive adjustment resistance (Yang, 2021). The F1 point is the main optimization criterion due to its relevance to unbalanced data sets. The randomized search yielded an optimized model that significantly outperformed the baseline, a configuration subsequently validated on a separate hold-out test set.

3.3.1.2 Explainability with SHAP and LIME

To complete the predictive performance, the SHAP and LIME, SHAP has been used on the XGBOOST model optimized, allowing double analysis of global and local predictions (Salih et al., 2023).
For the XGBoost model, the SHAP TreeExplainer was utilized. This specialized algorithm is computationally faster and provides exact Shapley values for tree-based models, avoiding the need for the slower, approximate sampling required by the model-agnostic KernelExplainer
LIME (local explanation of an understandable model): LIME has been applied to create local explanations for each student (Ribeiro et al., 2016). By adjusting an alternative model that can be understood around a selected body, LIME has revealed how specific functional values have led to a single prediction. This has provided local information, such as a unique VLE commitment file or demographic characteristics, enabling educators to use adapted information with personalized interventions (Nagy and Molontay, 2023). Along with each other, the double XGBOOST "Black Box" approach has been highly effective in a model that can interpret the ability to guide both institutional policies and personalized interventions.

3.3.2 Logistic Regression

Logistic regression (LR) I have included my project as an essential transparent model or a "white box" model. Its linear structure allows the coefficients of the model to be directly mapped to show the importance, which makes it better for my ideal explanation to compare with complex "black box" models.XGBOOST. LR is is set up to capture linear relationships in my dissertation, which makes it adapted to the prediction of dropout related to classification and continuous characteristics.

A basic logistics model is first formed with default parameters and is evaluated by using a 5-fold cross-validation. The evaluation data, including the accuracy, F1, and ROC-AUC, have been compiled with the priority of F1 to take into account the imbalance of the classes.
3.3.2.1 Super analysis adjustment

3.3.2.1 Hyperparameter Tuning

In order to improve the baseline model, the hyperparameter adjustment is made by a Grid search strategy. It focuses on resistance parameters (C), controlled in contrast to regularization applied to the model. The smaller values of C impose a more powerful form, preventing overfitting and adjusting the parameters by the models that are too complex, as larger values allow more flexibility but can reduce generalization at the other hand. The grid search was conducted with 5-fold stratified cross-validation, ensuring that class imbalance was preserved across folds and that performance estimates were robust. The objective of optimization is the F1 point, accurate balance, and recovery, and especially suitable for predicting "dropout". This has provided the optimal "c" value to provide the best compromise between deviations and variance. This model was later confirmed in a set of separate conservation tests to confirm generalization and avoid excessive overfitting.

3.3.2.2 Explainability with SHAP and LIME

LR served a dual role, also validating the dissertation's XAI framework for consistency of interpretation across models.

- Shap (explanation of Shaplley additives): LinearExplainer, specially designed for linear models, has been used to globally analyze the importance of functions. This technique provides the exact SHAP values that determine the contribution of each function The absolute means of the average SHAP of the results have been compiled in a list of influential variables. Above all, these shape results provide transparency by aligning with the model coefficients.
- LIME (Local Interpretable Model-Agnostic Explanations) then I created local explanations for cases of each student (Ribeiro et al., 2016). It works by adjusting simple alternative models and can be understood around specific predictions, revealing the unique level of combination of specific values that have affected a single result. The repetition of this in my analysis for different students (50,100,200)has proved how the risks of personalization appear from separate behavioral records and demographics, providing personal and individual information exploited by educators (Zarei and Pahl, 2025).

3.3.3 Random Forest

Random Forest was included as an ensemble-based benchmark model to provide a robust comparison with logistic regression and XGBoost, as complex, nonlinear interactions capture and overadaptive resistance in my dissertation . Despite its strong predictive performance, the ensemble structure poses a major challenge for accurate explanations, ultimately prioritizing XGBoost and logistic regression for deeper XAI analysis.

The fist thing the same I did with other two models i did with RF is first creating a baseline model trained on standard parameters and evaluated using 5-fold cross-validation.

Performance was assessed by accuracy, F1 score, and ROC-AUC to ensure comparability with other models.

3.3.3.1 Hyperparameter Tuning for Random Forest

Hyperparameter tuning is conducted using Randomized Search to efficiently explore a broad parameter space. With these values over the grid search included: n_estimators (50, 100, 200), max_depth (10, 20), min_samples_split (5), min_samples_leaf (10), and max_features (10). These parameters control model complexity and generalization. Optimization used 5-fold stratified cross-validation with F1-score as the metric, suitable for the imbalanced dropout dataset.

3.3.3.2 Explainability Challenges with SHAP and LIME

Explanability analysis with Shap and LIME showed limitations in Random forest averaging processes of individual characteristics, and making local prediction declarations less accurate (Salih et al., 2023). Although Computational inefficiency resulted in long-term and high memory consumption, the reliability of the approximate model declined compared to Xgboost and logistic regression, sometimes unstable and sometimes inconsistent explanations (Slack et al., 2019; Prolmper et al., 2023). Consequently, Random Forest remained an important criterion for performance assessment; it justified stronger compatibility between Xgboost and selected Xai technology with prioritization for detailed interpretability analysis, balancing knowledge generation and prediction accuracy (Nagy and Molontay, 2023).

3.3.4 Multi-Layer Perceptron (MLP): Deep Learning Component

The Multi-Layer Perceptron (MLP) represents the deep learning component of this project. This was chosen because of its ability to record highly nonlinear interactions and hierarchical properties, possibly revealing subtle patterns within OULAD datasets (Wang et al., 2024). Before training, functional standardization was used to ensure that all input variables contributed equally, improving numerical stability and promoting network convergence.

In contrast to tree-based models with optimized explanations, including MLP, neural networks with approval (Lundberg and Lee, 2017; Ribeiro et al., 2016) were also important to demonstrate the applicability of shape and LIME model expansion. MLP Baseline was implemented using Scikit-Learn's MLPClassifier using a single hidden layer, ReLU activation, and the Adam optimizer. Performance was assessed by accuracy, F1 score, and ROC-AUC using 5x layered cross-validation.

3.3.4.1 Hyperparameter Tuning

Hyperparameter tuning was conducted using a Randomized Search strategy to efficiently explore the large parameter space. The search included hidden_layer_sizes ranging from single-layer (50,100,500,1000,) to multi-layer architectures (128,64,32) to explore depth and capacity; activation functions {'relu','tanh'}; L2 regularization alpha from 1e−5 to 1e−2 to mitigate overfitting; and initial learning rates from 0.001 to 0.1. Three-fold stratified cross-validation prioritized F1-score to address class imbalance.

3.3.4.2 Explainability with SHAP and LIME

Shap and LIME were used for explanations to confirm the robustness of the framework conditions. Shap created a global characteristic ranking across medium absolute values using KernelExplainer in a representative subgroup (Lundberg and Lee, 2017). Meanwhile, LIME generated local explanations highlighting factors such as reduced VLE engagement (Ribeiro et al., 2016). This confirmed the applicability of both XAI methods to deep learning models (Salih et al., 2023). Nevertheless, MLP was ultimately excluded from the final implementation due to computing limitations. This highlights the compromise between the complexity of models and practical interpretability in research studies (Swamy et al., 2022).

3.4 XAI Implementation

The deployment of Xai has focused on logistics regression (LR) and Xgboost to approach the compromise of the basic explanation (Nagy and Molontay, 2023). LR, a transparent "white" model, is used as an important basis, providing direct understanding of the contribution of functions through coefficients and setting a reference for the quality of the explanation. XGBOOST, a highly effective "black box" model, has been chosen for outstanding prediction power in complex data (Yang, 2021). Applying forms (Lundberg and Lee, 2017) and LIME (Ribeiro et al This comparison approach proves the learning value of Xai.

3.4.1 Temporal explanation

Temporary explanation is an important component that provides dynamic information about students' behavior, which static models cannot grasp (Wang et al., 2024). The main method involves the formation of XGBOOST and Logistics regression models separated according to the weekly data taken from Oulad data (Kuzilek et al., 2017), lasting more than 26 weeks of the semester over 500 student.

This approach is important because the student's learning and factors affecting success are not static. By analyzing each week over the trained values created on both models, the development of the importance of time-out characteristics has been followed. The use of SHAP and LIME to fully explain. The implementation of the weekly formal analysis SHAP every week in which the average Shao values are calculated for each function in each weekly data frame (Lundberg and Lee, 2017).

These values have been stored in a Dataframe, creating a set of data on the importance of features over time. Similarly, the implementation of LIME has obeyed the temporary approach, creating and summarizing explanations for a representative of 50 students per week (Ribeiro et al., 2016).

Analysis of this dual method is very important to switch from explanation to action. The results will reveal what characteristics (such as early demographic data, VLE's participation, or midterm evaluation) become the most influential at specific points. This information is designed to shed light on interventions that are targeted and minimal, allowing educators to provide support when it best suits a student's learning path (Nagy and Molontay, 2023).

3.4.2 Fairness & Bias Auditing

Fairness & Bias Auditing as it is important and crucial for potential biases in a high-stakes domain field such as education . A comprehensive fairness and bias-audit of equity and bias has been carried out on XGBOOST and Logistics regression models that use both shapes and LIME. The approach involves a systematic analysis of the behavior of the model through specific demographic groups.

The goal is intentionally placed on three important sensitive attributes recognized in the study of equity: gender, age, and disability. This allows direct inspection of moral concerns. This method is systematic as it involves repeating for each value of demographic characteristics to create different small groups for testing. For each group, the explanations of Shap and LIME have been calculated to understand local predictions. Especially valuable is the ability to create understandable explanations, which is the key to conveying potential biases to other parties involved in it(Swamy et al., 2022).

The basic analysis includes the synthesis and visualization of the values of SHAP and LIME for each subgroup. This comparative step has allowed me to directly evaluate the importance of the characteristics and the average contribution of the characteristics between different demographic groups (for example, to determine whether the age range is not adequate for the prediction of a group for other groups). This dual approach is essential to determine the unmatched potential difference thanks to the global measures of the model.

3.4.3 Explanation Robustness

Evaluating the reliability of the explanations is a significant consideration in Xai, because unstable explanations change over drastically with unnecessarily small input perturbations for highly problematic applications (Prümper et al., 2023). To invetigate this, i did a systematic and detailed analysis on XGBOOST and logistic regression models with shape and LIME.

The method started by selecting a student representative of the test data set to create an initial explanation. A disturbed data version was later created by introducing a minimum actual change: an increase of 5% for the Total_Clicks function, a major behavior variable. This isolated test allows a clear assessment of stability in the context of reasonable changes in student activity (Oluwatobi, 2025).

The basic analysis is to directly compare the initial explanations and is seen in the results chapter, focusing on the stability of classification of the importance of the characteristics and amplitude of their contributions. A powerful explanation will keep the most important influential characteristics, while an unstable one will make significant classification changes (Slack et al., 2019). By rigorously testing the stability of the explanations, the analysis provided a critical evaluation of whether the models' rationales hold up under slight changes in student behavior, adding a crucial layer of academic rigor regarding XAI reliability.

3.4.4 Explanation Clustering

The final step in the XAI methodology was to translate the individual explanations into the educator's implementable knowledge and move them to a practical, policy-driven solution. This was achieved by applying K-Means clusters to the Shap and LIME value vectors of student data and grouping students with similar predictions (Cooper, 2022).

The systematic methodology involved a comprehensive explanation of calculations. Shap values for the entire test set (shap.treeexplainer for xgboost and shape.linearexplainer) and LIME declarations for representative samples of 500 students.

The K-means algorithm was applied directly to a matrix of these explanatory values. nclusters = 3 was used to present three different student individuals. This provides a balance between grouping and interpretability. Each cluster was characterized by analyzing the average characteristic meaning (mean absolute Shap/LIME values) within this group. By defining these distinct personas (e.g., "assessment-driven students"), methodologies effectively combine the complex behaviors of the model with practical educational frameworks for target support and intervention (Zarei and Pahl, 2025).

Summary for Chapter 3

This chapter defines a robust methodological framework for research. After systematically assessing four different models (XGBoost, Random Forest, MLP, and Logistic Regression) with cross-validation and hyperparameter tuning, XGBOOST and logistic regression were chosen for XAI implementation due to their superior prediction accuracy. This justifies the core comparative analysis. XGBoost acts as a powerful "black box" model, and logistic regression serves as an interpretable "white box" benchmark. The next chapter presents comprehensive SHAP and LIME outcomes covering the knowledge of temporal, fairness, robustness, and clustering that arise from these two selected models.

Chapter 4: Results

This chapter offers an overview of the research results, answering research questions. It starts with EDA insights, followed by comparing four prediction models, highlighting the compromise between complexity and accuracy. This chapter emphasizes XAI analysis, including global and local explanations for XGBOOST and Logistics regression, dynamic time information, fair and biased audit, and strong inspection. Finally, it presents the main characteristics of the characters of separate students originating from the group's explanation, connecting technical results with educational information that can be used.

4.1 Exploratory Data Analysis (EDA) and Data Rationale

The initial Exploratory Data Analysis (EDA), presented in full detail in Appendix A.1, was crucial for understanding the OULAD dataset structure and informing the final model selection strategy.

- Target Variable Imbalance: The EDA confirmed a significant class imbalance in the final target variable, with approximately of students successfully completing the course (Pass/Distinction) versus 25% who dropped out (Fail/Withdrawn) (Figure A.1.5). This imbalance justified the methodological choice to prioritize F1-Score and ROC-AUC over simple accuracy as the primary optimization and evaluation metrics.
- Feature Dynamics: Visualizations confirmed the highly dynamic, non-uniform nature of student engagement over time (Figure A.1.1), directly validating the need for the advanced temporal explanation analysis performed later in this chapter.
- Fairness Baseline: The EDA established a baseline of fairness by showing no significant linear differences in VLE engagement between certain demographic groups (Figure A.1.3), setting the stage for the rigorous XAI audit to uncover subtle, non-linear biases that descriptive statistics cannot detect.

4.2 Model Performance

This section presents a comparative analysis of predictive performance for all four models, from baseline to hyperparameter-type conditions. The purpose of this analysis is not only to demonstrate a compromise between model complexity and predictive power, but also to justify the choice of two most effective models, XGBoost and logistic regression, for subsequent XAI implementations.

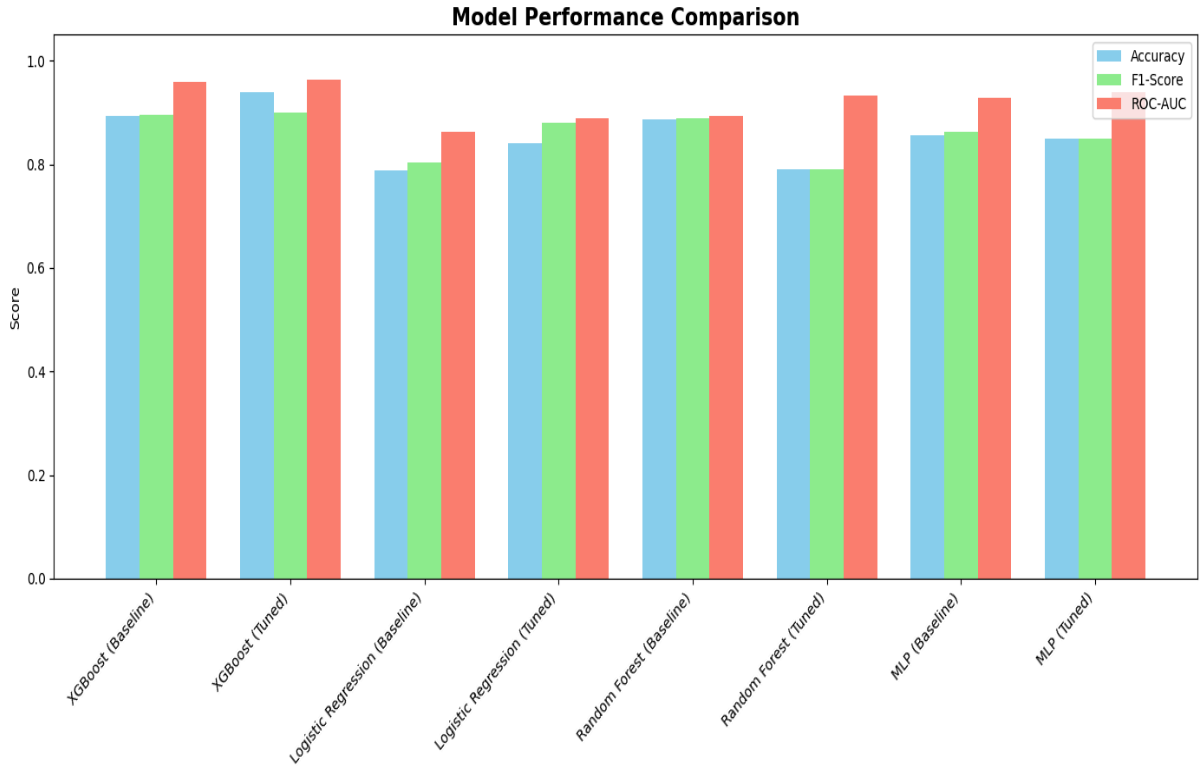


Figure 3: A bar chart visually comparing the performance metrics (Accuracy, F1-Score, and ROC-AUC) of all four models at both baseline and hyperparameter-tuned stages.

Figure 3 shows a clear visualization of the prediction values for each model. X-axis, baseline model, and tuning status classify the performance of each model as a baseline or tuned version. Scores, which are the Y-axis, quantified performance with three important rating metrics: accuracy (light blue), F1 score (light green), and ROC-AUC (salmon). This number highlights that all models have strong basic performance, but emphasizes that hyperparameter tuning leads to a clear increase in performance, especially in the ROC-AUC scores of top-performing models. Based on these robust performance metrics, XGBoost and logistic regression were selected as the two main models for incoming XAI analysis. This decision is twice as high. XGBoost's excellent performance makes it an ideal "black box", and the strong performance and inherent interpretability of logistic regression provide a definitive "white box" for verification and comparison of XAI knowledge.

4.3 Global Feature Importance: XGBoost vs. LR

This part directly compares the importance of the global characteristics of the two main models, XGBOOST and logistic regression, providing an important link between the complexity of the model and the ability to interpret.

4.3.1.SHAP XAI Implementation on XGBoost vs. Logistic Regression

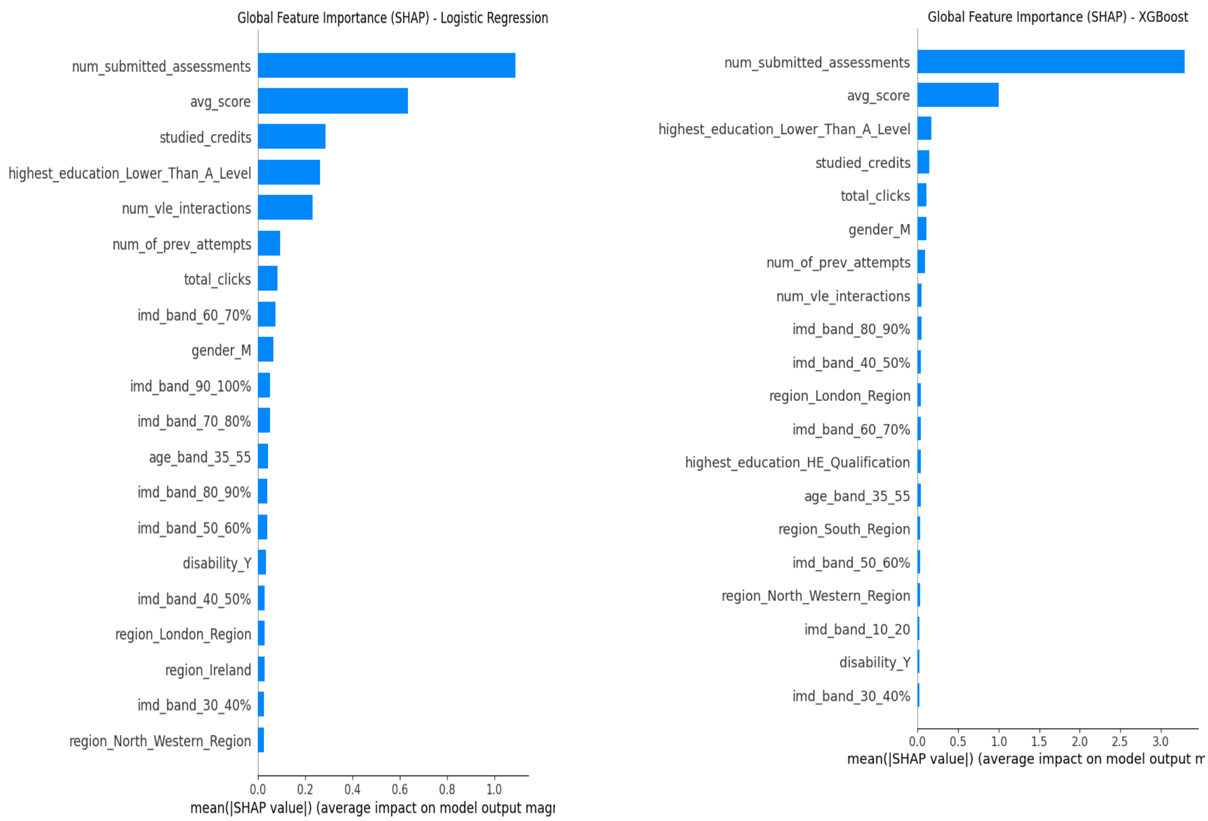


Figure 4: A bar plot comparing the global feature importance of the XGBoost and Logistic Regression models, ranked by the mean absolute SHAP value.

The layout of the bar chart in Figure 4 has a clear contrast between the two models. The y-axis lists the features, while the x-axis, "mean(|SHAP value|)," quantifies their average impact. The logistics regression model shows the importance of more dispersed and less pronounced functions, with 'num_submitted_assessments' and 'AVG_Score' being the first two features. On the other hand, the XGBOOST model shows a much greater dependence on these two characteristics; their impact is greater in order than any other feature. The main conclusion is that, although the two models determined the most important features, their dependence level on them is significant. XGBoost, as a strong overall model, learns a much more complex relationship, providing much more weight for some main predictions. This justified the "black box" label because its high performance is motivated by some dominant factors. On the other hand, the logistics regression model, although less accurate, provides a more balanced view of the importance of the characteristics, in which a wider range of functions contribute more evenly to the final prediction. This comparison is very important for the project because it provides a stable illustration of the compromise between the performance of the model and the simplicity of its internal logic.

4.2.2 Local explanation with LIME (XGBOOST)

Although global explanations provide a high-level overview of model behavior, local explanations are essential to understand why a specific prediction is made for an individual. This section presents an analysis of LIME about the prediction of a representative student, both the logistics and XGBOOST models, revealing a unique combination of characteristics that affect their expected results. The analysis is presented to students in Index 200, which will be "abandoning" with the great confidence of the two models.

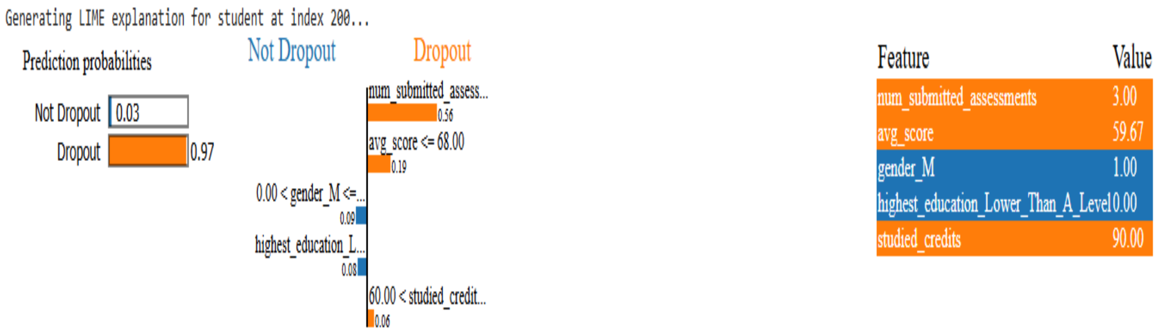


Figure 5: The LIME explanation for the XGBoost model's high-confidence "Dropout" prediction for student 200, highlighting the key features and their contributions.

Explanation of LIME in Figure 5 for the high -performance XGBOOST model shows a more complex and more complicated decision implementation process. As pointed out in figure 5, while the model also relies on a low number of submitted assessments and a low average score, the explanation highlights a crucial finding: a demographic feature, gender_M, also contributes to the prediction.. This shows the maximum sensitivity of the model with appropriate interactions and provides an important overview that an educator can use for a timely intervention. The comparison of LIME outputs for the same student of the two models is very significant because it provides a tangible illustration of the compromise between the performance of the model and the complexity of its internal features

4.2.3 Local explanation with LIME (Logistic Regression)

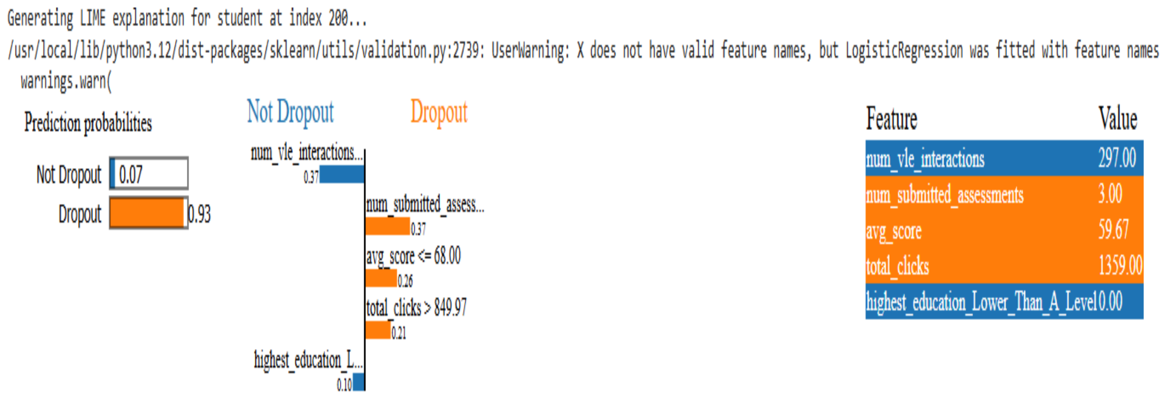


Figure 6 : The LIME explanation for the Logistic Regression model's high-confidence "Dropout" prediction for student 200, showing the key feature contributions.

The LIME analysis for the Logistic Regression model, our interpretable baseline, provides a clear and intuitive breakdown of the student's prediction.. The explanation for the predictions of high reliability "drop" (probability of 0.93) is strongly attributed to the combination of characteristics. As seen in Figure 6 low VLE interactions (num_vle_interactions), low number of ratings submitted, and low mean scores are the main drivers of negative outcomes. This shows that logistic regression models are predictable, easy to interpret, and provide a local explanation that greatly reflects the direct linear relationships of the data

4.4 Temporal Explanations

This section presents the results of a temporal explanatory analysis that represent the critical elements of this project. The purpose of this analysis is to demonstrate that predictive justification of the model is not static, but a dynamic process that develops along with student journeys. These results directly address the research question of how the importance of characteristics over the course of the semester shifts due to the strong XGBoost model.

4.4.1 XGBoost: Dynamic Shift in Feature Importance (SHAP and LIME)

The systematic shift in the predictive focus of the XGBoost model is visually demonstrated in Figure 7, which quantifies feature influence using Mean Absolute SHAP values. This line plot tracks the time progression over 26 weeks (x-axis), with the average effect of features on the model's output quantified on the y-axis.

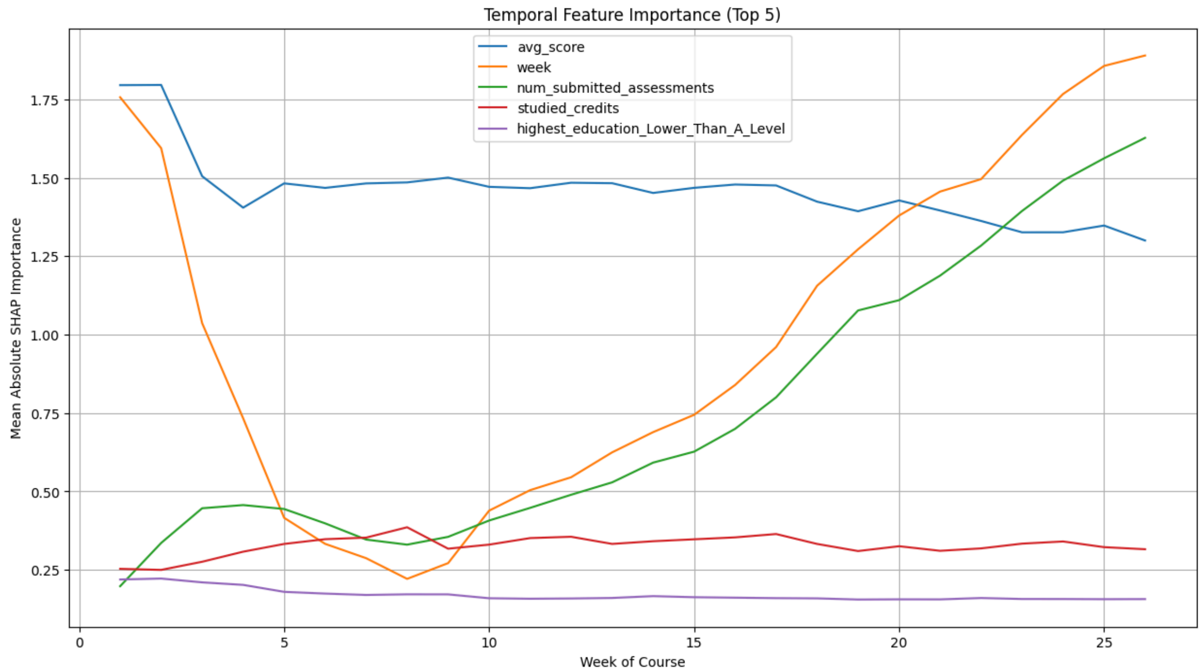


Figure 7: A line plot illustrating the temporal evolution of the top 5 most important features for the XGBoost model, as ranked by their mean absolute SHAP value over 26 weeks.

In the Early Stages (Weeks 1-5), the model is primarily influenced by generalized indicators, such as the week proxy itself and immutable background factors like prior education features. This reliance on demographic and proxy data is a consequence of limited behavioral and academic history being available in the early semester.

A highly significant and actionable shift occurs in the Mid-to-Late Stages (Weeks 6-26) as direct academic and engagement features gain dominance. Evaluation-related characteristics, specifically AVG_SCORE and num_submitted_assessments, rapidly become the most influential predictors. This finding confirms that formal performance and successful assessment completion become the most reliable indicators of the final outcome in the latter half of the semester.

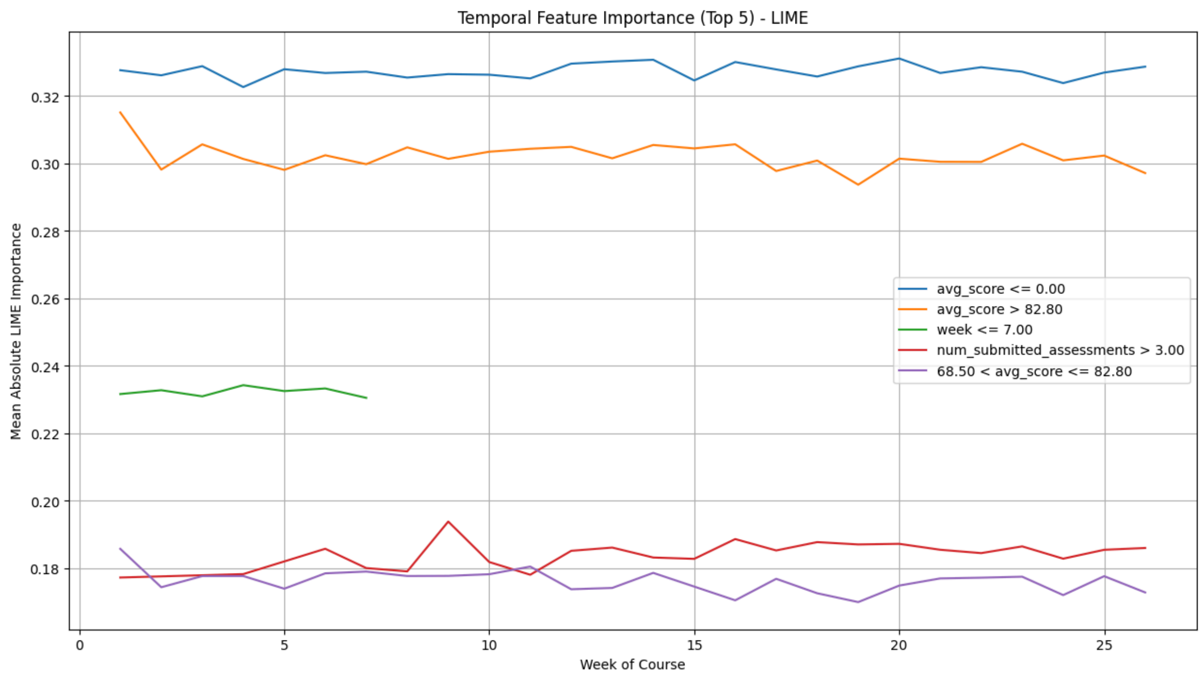


Figure 8: A line plot illustrating the temporal evolution of the top 5 most important features for the XGBoost model, as ranked by their mean absolute LIME value over 26 weeks.

The temporal analysis using LIME (Figure 8) strongly confirms this dynamic trend for the black-box model. The LIME outputs show a consistent pattern where early reliance on general commitment gives way to the clear control of assessment-related features later in the course. This dual-method validation significantly enhances the reliability of the observed temporal trends for the XGBoost model.

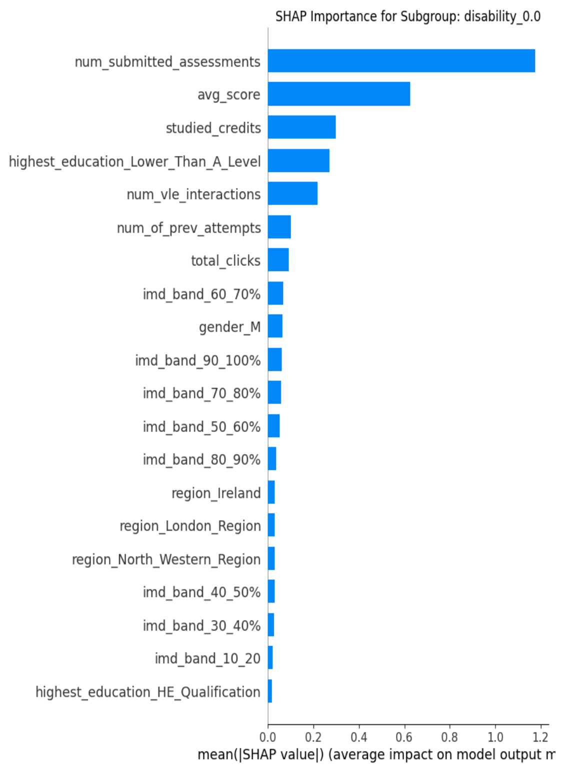
4.4.2 Logistic Regression: Stability as a White-Box Benchmark

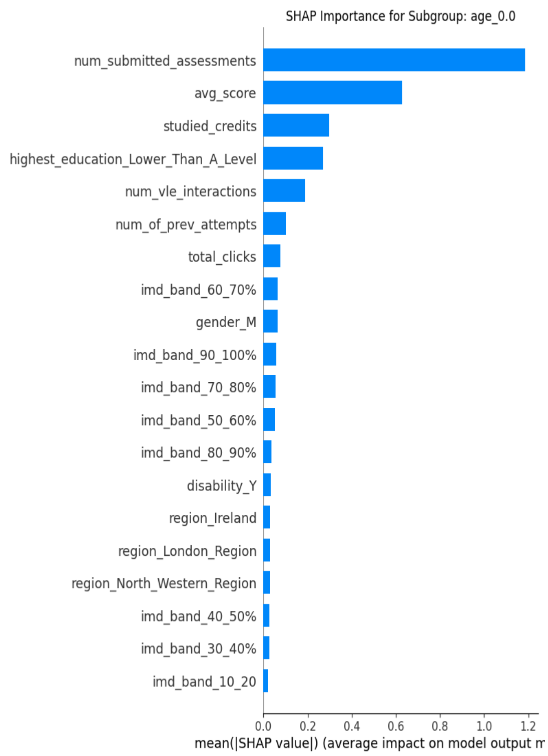
In contrast to the dynamic XGBoost model, the Logistic Regression (LR) model displays a high degree of stability in feature importance across the 26 weeks, as detailed in Appendix A.2.1 (SHAP) and A.2.2 (LIME).

For the LR model, core assessment features, particularly num_submitted_assessments, remain highly influential throughout the entire semester, showing a near-linear, consistent relationship with the dropout outcome. This stability is not a failure of the LR model but a successful validation of its intrinsic linearity and transparency. The model's rigid linear structure prevents it from learning the complex, non-linear, time-dependent interactions that drive the XGBoost's dynamic feature usage, reinforcing its role as a stable benchmark against which the dynamism of the complex black-box model is measured.

4.5 Fairness & Bias Audit Results

This section presents the results of the fairness and bias audit, and details how the prediction of models changes in other small groups of the main demographic groups. The purpose of this analysis is to directly confront ethical considerations and to ensure that the models are not affected by the sensitive characteristics. My analysis focuses on the contribution of gender, age, and disability to logistic regression and XGBOOST (Appendix) mode





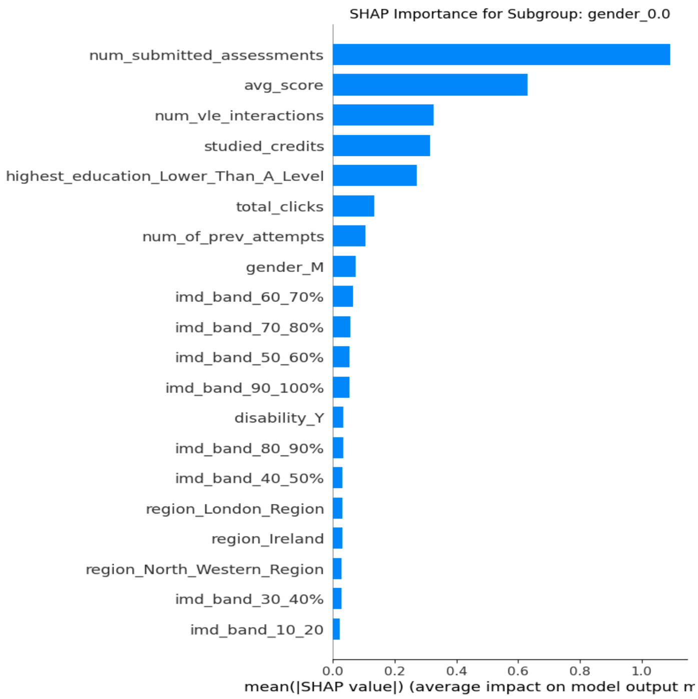


Figure 9: Top 10 Features by Mean Absolute SHAP Value (Logistic Regression)

The audit is done by comparing the average values of the absolute SHAP for each subgroup, revealing the potential difference in the way models are based on characteristics. The results of the logistics regression model, indicated in the following figure 9 , indicate that if the most coherent characteristics between gender, age, and disabilities, the relative importance of other characteristics may vary. For example, the two main features of all sub-groups are number_summed_ssssments and AVG_Score, which is the result of expected and fair. However, more testing of results shows an important discovery for the Logistic regression model (as shown in Figure 9), in which the gender contribution is an important factor in some personal predictions.

4.5.1 Logistic Regression: The Fair Benchmark

The audit of the transparent Logistic Regression (LR) model provided a necessary benchmark for fairness. As a linear model, the LR's feature importance (both global and local, documented in Appendix A.3.2) is heavily concentrated on core academic and behavioral features. This finding confirms the model's fundamental fairness: its predictive logic relies on easily interpretable and directly actionable student effort metrics (like `num_submitted_assessments` and `AVG_Score`). Crucially, the LR model consistently showed minimal reliance on sensitive demographic attributes in local predictions, confirming its rigid, linear structure successfully avoids learning diffused, localized biases present in the data.

4.5.2 XGBoost: Evidence of Diffused Bias

In contrast, the audit of the high-performance XGBoost model reveals a more nuanced and concerning pattern of feature usage. This analysis leveraged local-level LIME outputs across demographic subgroups (Figure A.3.1, Table A.3.1).

While the academic features (submissions and score) remain the primary global drivers, local analyses, such as those presented in Table A.3.1, indicate a continuous, non-zero reliance on sensitive attributes:

- Subgroup Influence: Across male, female, and non-disabled subgroups, the sensitive feature condition `gender_M` constantly appears within the top four or five contributors to local predictions.
- Significance: This inclusion, while small in magnitude, confirms that the complex, non-linear architecture of the XGBoost model is capable of leveraging subtle correlational signals from sensitive attributes.

This evidence of localized reliance on demographics is critical. It proves that the XGBoost model, while achieving high accuracy, carries the potential for algorithmic bias at the individual prediction level, an issue that would be entirely masked by standard global performance metrics. This highlights the necessity of using XAI techniques to audit for bias before deployment, ensuring equitable outcomes across all student demographics.

4.6 Explanation Robustness:

This section presents the empirical results of the perturbation analysis, a critical methodological step to assess the reliability of the model explanations. The analysis directly addresses Research Question 5 by comparing the stability of SHAP and LIME explanations for the XGBoost model when subjected to minor data perturbation (a increase in the Total_Clicks feature). The stability results for the Logistic Regression benchmark are provided in Appendix A.4.

```
Analyzing explanation for student at index 50...
Original Prediction: Dropout

--- Original SHAP Explanation (Top 5) ---
                                 Feature  SHAP_Value
5             num_submitted_assessments    5.753392
4                             avg_score    1.261304
0                    num_of_prev_attempts    0.876879
30                        imd_band_80_90%    0.162874
20  highest_education_Lower_Than_A_Level    0.136934

Perturbed Prediction: Dropout

--- Perturbed SHAP Explanation (Top 5) ---
                      Feature  SHAP_Value
5   num_submitted_assessments    5.775480
4                   avg_score    1.178300
0        num_of_prev_attempts    0.876483
2                total_clicks    0.185516
30            imd_band_80_90%    0.162874

--- Comparison of Explanations ---
Original top 5 features:
                                 Feature  SHAP_Value
5             num_submitted_assessments    5.753392
4                             avg_score    1.261304
0                    num_of_prev_attempts    0.876879
30                        imd_band_80_90%    0.162874
20  highest_education_Lower_Than_A_Level    0.136934

Perturbed top 5 features:
                      Feature  SHAP_Value
5   num_submitted_assessments    5.775480
4                   avg_score    1.178300
0        num_of_prev_attempts    0.876483
2                total_clicks    0.185516
30            imd_band_80_90%    0.162874
```

Figure 10: Comparison of SHAP explanations for Student 50 before and after a perturbation of the total_clicks feature, demonstrating the stability of feature ranking and predictive rationale in the XGBoost model.

An important factor in my research is to assess the reliability of the explanation. This is a significant consideration in XAI, because an unstable and completely changed explanation with small entry disorders is not reliable for highly problematic applications. To study this, I made a disturbing analysis on XGBOOST and logistics regression models that use both shapes and LIME.

Figure 10: Compare explanations about the shape of 50 students before and after the Total_Clicks function disturbance, showing the stability of the functional classification and the prediction of the XGBOOST model. Figure 10 presents a direct comparison and interpretation of each other about the SHAP of a single representative student (Index 50) before and after an increase of 5% compared to the Total_Clicks function. Its objective is to test the strong experiment of the model's justification, thus confirming the reliability of the method of SHAP will be used in the context of education. The prediction of the origin has been droupout, and this prediction remains the same after the interruption. Outputs display two lists of 5 main features with their corresponding shap_value. Explain the initial SHAP showing the justification of the model based on students' actual data. Explaining the disturbing SHAP shows the justification of the model after minor change has been included in Tthe otal_Clicks function. Classification of features - Num_summetted_asssides (values of the original shape: 5.75), AVG_Score (1.26), retained their classification and is the main pilot of the "selected" prediction in two scenarios. The oscillation of the value is observed while the classification is stable; the degree of the formal values has changed a bit, as shown by AVG_Score (1,261304 → 1,178300), showing the maximum quantitative change. It confirmed that the Xgboost model is not fragile; Its basic anticipation is anchored in the learning performance (submitted and score) and is still consistent even when small noise is included in behavioral data.

This analysis confirmed that the formal explanation created for the XGBOOST model is stable and reliable. This stability confirms the method used in this project and brings great confidence that ideas come from formal analysis, such as temporary and group analysis, which is reliable to clarify exploitation interventions.

```
Analyzing explanation for student at index 50...
Original Prediction: Dropout

--- Original LIME Explanation (Top 5) ---
                                              Feature  Contribution
0                             total_clicks <= 849.97      0.116127
1                         68.00 < avg_score <= 72.77     -0.112061
2                            0.00 < gender_M <= 1.00     -0.110874
3             4.00 < num_submitted_assessments <= 6.73     -0.101182
4  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.048930

Perturbed Prediction: Dropout

--- Perturbed LIME Explanation (Top 5) ---
                                              Feature  Contribution
0             4.00 < num_submitted_assessments <= 6.73     -0.119512
1                            0.00 < gender_M <= 1.00     -0.116923
2                         68.00 < avg_score <= 72.77     -0.090462
3                            studied_credits > 120.00      0.087554
4  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.058527

--- Comparison of Explanations ---
Original LIME Explanation:
                                              Feature  Contribution
0                             total_clicks <= 849.97      0.116127
1                         68.00 < avg_score <= 72.77     -0.112061
2                            0.00 < gender_M <= 1.00     -0.110874
3             4.00 < num_submitted_assessments <= 6.73     -0.101182
4  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.048930

Perturbed LIME Explanation:
                                              Feature  Contribution
0             4.00 < num_submitted_assessments <= 6.73     -0.119512
1                            0.00 < gender_M <= 1.00     -0.116923
2                         68.00 < avg_score <= 72.77     -0.090462
3                            studied_credits > 120.00      0.087554
4  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.058527
```

Figure 11 Comparison of LIME explanations for Student 50 before and after a perturbation of the total_clicks feature, demonstrating the significant instability and shift in feature contribution within the LIME local approximation model.

In this figure 11, a direct comparison of the LIME declarations of one representative student (index 50) is displayed side by side after increasing the Total_Clicks feature. The aim is to empirically test the robustness of the model's justification and to verify the reliability of the LIME method for use in educational contexts. . The original prediction was to see how many students would drop out, and it stayed unchanged even after the failure. The outcomes display the unique LIME description in addition to the obstructed LIME statement for the pinnacle 5 attributes. The Function column presentations the variety of function values (e.g., total_clicks <= 849.97), while the 'Contribution' column indicates the quantity and direction (positive or negative) of each feature's impact on prediction. The perturbed explanation fully removes the original top feature, total_clicks <= 849.97 (Original Contribution: 0.116127). This shows that a mild change in a single feature value was sufficient to absolutely adjust the version`s perceived nearby dependency.Change in Magnitude in Significant adjustments in rating and contribution also are proven in capabilities like gender_M and avg_score, demonstrating how touchy LIME's nearby approximation is to enter noise. This instability increases the opportunity that the intricate, non-linear choice obstacles of the XGBoost version can be hard for LIME's nearby approximation approach to as it should be comprehend. When motives are wanted for high-stakes treatments, this locating acts as a critical warning, highlighting the need of the use of theoretically primarily based totally strategies like SHAP..

4.7 Explanation Clustering and Student Personas

This section presents the results of the explanations for the clustering analysis. This is an important step in transforming model insights to individual levels and becomes a practical framework for educators. The main purpose of this analysis is to identify different groups of students based on predictive rationality. This creates personas that can affect targeted intervention strategies.

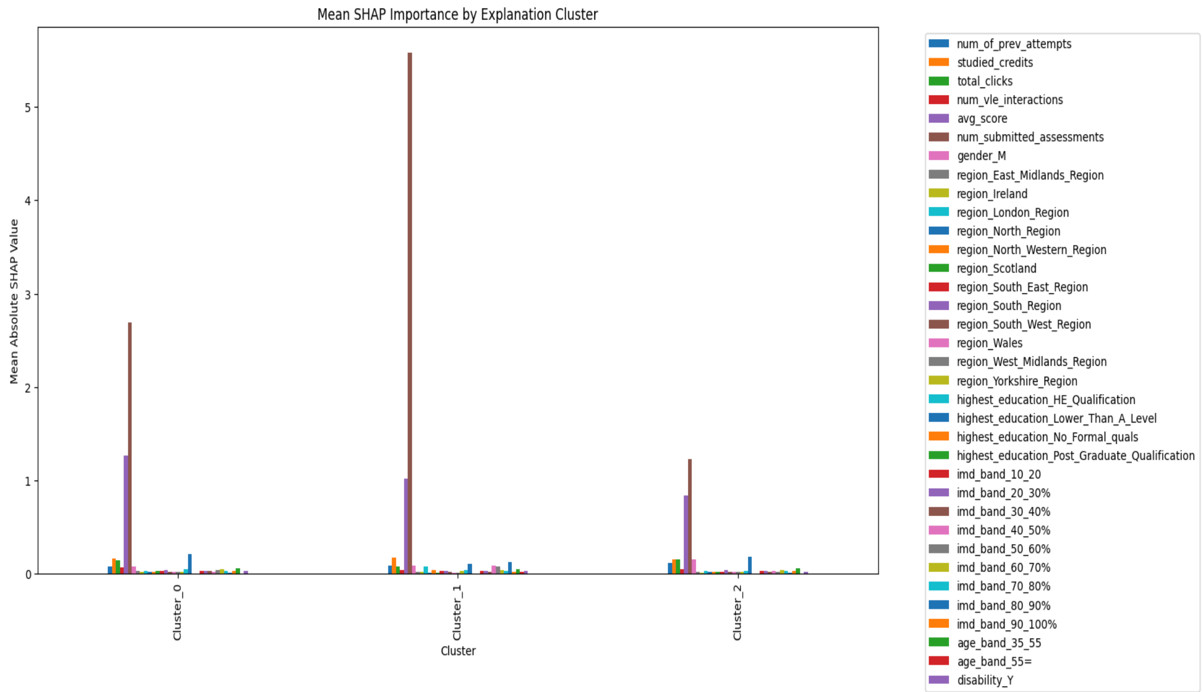


Figure 12: A bar chart illustrating the mean SHAP importance for each of the three identified student explanation clusters.

Figure 12 illustrates the Results of a medium K-means with SHAP values. The X-axis shows the three clusters identified (clusters 0, 1, 2), and the "medium absolute SHAP value" on the y-axis quantifies the average effect of each characteristic within this cluster. Each bar represents a different function of the legendary display method. The analysis of this diagram allows for the identification of three different students.

- Cluster 0 (Evaluation - Driven Student): This cluster is characterized by a high dependency on num_submondd_assessments and avg_score as the main drivers of model prediction. These students have a strong commitment, but are ultimately evaluated based on academic achievement.
- Cluster 1 (VLE Engagement Student): This cluster is defined by its extreme importance for VLE engagement functions such as Total_Clicks and Num_VLE_interactions. This indicates that this group is the most important factor in the predicted outcomes of this group.
- Cluster 2 (demographic and previous experience-driven students): This cluster shows that it relies heavily on demographic characteristics, such as highest_education_Lower_Than_A_Level, and studied_credits. This indicates that for this group, their prior academic history and background are the most influential factors in their predicted outcome.

This clustering analysis is extremely important as it combines the complex behavior of the model with a practical educational framework. Identified personas provide a clear roadmap for intervention. This indicates that different groups of students may need different types of support. This methodology transforms the model into a concrete, implementable strategy, and is explained in more detail in the next chapter. This chapter provides a comprehensive overview of the findings that address all research questions.

This Chapter started with exploratory data analysis, and lesson imbalances highlight and justify robust performance metrics. The model evaluation showed that Xgboost and logistics regression achieved the highest prediction accuracy and supported the selection of incoming XAI analysis. Global and local explanations showed significant predictive characteristics, while time explanations pointed to the later deferral characteristics due to demographic factors at the onset of academic achievement. The fairness audit confirmed the stability of student explanations grouped with potential local bias, robust testing, and personas that can be implemented to check practical interventions and knowledge of linked models.

Chapter 5

This Chapter goes beyond objective results to provide an important interpretation of the importance of student drop prediction. This argument integrates important findings of compromises in the interpretability of accuracy, explains dynamic changes in characteristic meanings, and explains the importance of identified student personas. This narrative of this chapter translates technical results into understanding, answers research questions, and deepens understanding of the learning process recorded by Ooulad data records (Kuzilek et al., 2017).

5.1 Findings

5.1.1 Model Performance (RQ1)

Based on model performance outcomes, multi-model methodology successfully identifies key elements of student success within the OULAD dataset using the unique strengths of both interpretable and efficient models (Lopez et al., 2024). The powerful model XGBoost is essential for determining the maximum predictive (ROC-AUC 0.9628, F1 score 0.90). The excellent performance confirms the model's ability to grasp the most important factors that drive student success: complex, nonlinear characteristic and feature interactions. The effectiveness of the Random Forest quadratic ensemble model (ROC-AUC 0.9624, F1 score 0.89) continues to test that complex, tree-based methods are needed to fully model these successful factors (Waheed et al., 2022).

This powerful performance is consistent with the results of other research findings on the dates of educational data (Yang, 2021; Fernandes et al., 2020). It is important that the interpretable model, logistic regression (LR) (ROC-AUC 0.8683, F1 score 0.81), serve as the required interpretability factor. The solid performance of the LR shows that it is less accurate, indicating that the most important element of the student, the core predictive signal, is recorded by linear relationships. Multi-model comparisons allow for the final identification of key elements. Factors that are linearly visible in the core (verified by LR and amplified by XGBoost) are most important, but performance from 10-15% represents the added value of complex, nonlinear features.

By using Xai for both (Ribeiro et al., 2016; Lundberg and Lee, 2017), the paper moves beyond simple predictions and explains how and why these complex factors (xgboost) or simple factors (LRs) lead to success.

5.1.2 Global and Local Description (RQ2)

This section significantly interprets the main XaI findings, showing the critical difference in the interpretability of the high-performance black box model (xgboost) compared to the transparent white box model (logistics regression).

Global Feature Importance using Shap (Lundberg and Lee, 2017) presents a fundamental contrast to how all models learn across data records, clearly indicating a compromise in the interpretability of accuracy. As shown in Table 1, both models unanimously identify num_submitted_assessments and avg_score as the two most influential features. However, the XgBoost model places exponentially high meanings as logical regressions (1.09 and 0.63, respectively) for these two best features (3.29 or 0.99). This shows that the performance of the black box model is achieved by learning more complex and nonlinear relationships that build enormous weights due to the statements of dominant academic predictors (Yang, 2021). In contrast, the LR model further distributes importance over a wider range of characteristics, reflecting simpler and linear logic.

Table 1: Global Feature Importance Comparison (SHAP)

Rank	Logistic Regression Feature	Importance	XGBoost Feature	Importance
1	num_submitted_assessments	1.09	num_submitted_assessments	3.29
2	avg_score	0.63	avg_score	0.99
3	num_vle_interactions	0.33	highest_education_Lower_Than_A_Level	0.16
4	studied_credits	0.31	studied_credits	0.14
5	highest_education_Lower_Than_A_Level	0.27	total_clicks	0.11

LIME's local explanation for representative students (Ribeiro et al., 2016) (Index 200) further clarifies this difference and shifts the focus to individual predictions. The contributions of local features are shown in Table 2. Calculating the analysis of logistic regression models. This will strongly assign the explanation to expected factors such as num_vle_interactions and num_submant_assesms. This confirms that LR inference is transparent and highly reflects the direct linear relationship of the data. Conversely, the LIME declaration in the XGBoost model shows a critical value. Xgboost is dominated by academic performance (Num_Submitted_Assessment of 0.56), while the demographic function gender_m has been featured as a key local participant (0.09). This subtle inclusion, which was not a top-five factor in linear LR, indicates that black box models are more sensitive and capable of finding nonlinear signals with sensitive attributes. This finding proves that highly accurate models are simultaneously more susceptible to subtle properties and that the need to use Xai not only for general understanding, but also for ensuring fairness for granular inspection (K.V. S. et al. , 2025).

Table 2: Local explanations Feature Importance Comparison (LIME)

Rank	Logistic Regression Feature	Importance	XGBoost Feature	Importance
1	num_vle_interactions	0.37	num_submitted_assessments	0.56
2	num_submitted_assessments	0.37	avg_score≤68.00	0.19
3	avg_score≤68.00	0.26	gender_M	0.09
4	total_clicks>849.97	0.21	highest_education_L ...	0.08

5	highest_education_L...	0.10	60.00<studied_credits≤90.00	0.06

5.1.3 Temporal Explainability (RQ3)

The temporal explanatory analysis proved decisive, revealing clear and systematic changes in the predictive focus of the XGBoost model over the 26-week course, thereby directly validating the hypothesis that student trajectories are dynamic processes (Wang et al., 2024).

5.1.3.1 Dynamic Feature Evolution and Intervention Timing

This finding forms the basis of the argument that static interpretation is insufficient for effective intervention. Analysis based on SHAP values (Figure 7) provides a powerful, actionable narrative of this dynamic behaviour:

- Early Intervention Window (Weeks 1-4): Predictive factors are dominated by demographic characteristics and proxies for initial engagement. This suggests early interventions must be pastoral and commitment-focused (e.g., ensuring access to materials, promoting engagement skills) before academic failure occurs.
- Mid-Semester Shift (Weeks 5-18): A significant shift occurs as model focus moves to direct academic feedback (avg_score). This signals the crucial need to prioritise learning support and skill development during the mid-semester, before assessment failures become irreversible.
- Late-Stage Predictors (Weeks 19-26): Assessment-related features become overwhelmingly influential. Late-stage support must focus on mitigating the cumulative effects of disengagement or poor performance.

5.1.3.2 Comparative Validation

The dynamic evolution observed in the black-box XGBoost model was robustly validated by the LIME analysis (Figure 8). Crucially, the transparent Logistic Regression model (Appendix A.2.1, A.2.2) showed no such dynamic shift, maintaining a near-constant reliance on core assessment features. This contrast confirms two critical points: first, the dynamic evolution is a genuine phenomenon captured by the powerful XGBoost; and second, the LR model's inherent stability confirms its rigid linear structure, which is incapable of learning the subtle, time-dependent interactions that drive the optimal intervention timing (Nagy and Molontay, 2023).

These results provide a powerful argument to move beyond static predictive models. Educators can now be informed that support should be staged and tailored to the most influential factors at key intersections, providing a direct connection between model insights and human-centered solutions.

5.1.4 Bias and Fairness Audit (RQ 4)

This section interprets the decisive findings from fairness and bias and shows how Xai technology has potential algorithmic bias and differences in sensitive demographic subgroups, including gender, age band, and disability status.

Testing the transparent Logistic Regression (LR) model provided a fundamental benchmark for ethical performance. The LR's SHAP analysis (Table 3) showed that the top features driving predictions across all subgroups were consistently linked to academic performance (num_submitted_assessments and avg_score). This domination by actionable academic signals confirms the LR model's inherent fairness; its rigid, linear structure prevents it from learning the subtle demographic correlations that lead to bias.

Table 3: Top 10 Features by Mean Absolute SHAP Value (Logistic Regression)

Subgroup	Rank	Feature	Importance
Gender (Female)	1	num_submitted_assessments	1.09
	2	avg_score	0.63
	3	num_vle_interactions	0.33

	4	studied_credits	0.31
Age (Under 35)	1	num_submitted_assessments	1.19
	2	avg_score	0.63
	3	studied_credits	0.30
	4	highest_education_Lower_Than_A_Level	0.27
Disability (No)	1	num_submitted_assessments	1.18
	2	avg_score	0.63
	3	studied_credits	0.30
	4	highest_education_Lower_Than_A_Level	0.27

The XGBoost model audit, however, revealed a more concerning pattern through granular, local-level LIME analysis (Ribeiro et al., 2016). While the model's global decision drivers remained academic, local analyses (Table A.3.1 in Appendix A.3) showed that sensitive attributes could be a significant local factor in individual student predictions. The consistent, non-zero reliance on features like the `gender_M` condition in the local top-five explanations is evidence of the black-box model's ability to leverage subtle demographic signals more effectively than the simple LR model.

This localized bias is critical to ensuring ethical and responsible AI. The results demonstrate the necessity of moving beyond simple global fairness metrics to uncover the potential for disparate impact. The granular XAI audit mandates that data scientists must not only achieve accuracy but also perform this type of local inspection to ensure that the model is fair and equitable for all students (K. V. S. et al., 2025). This systematic approach confirms that high-stakes predictive systems require continuous monitoring and debiasing before deployment.

5.1.5 Explanation Robustness (RQ5)

This section interprets the results of the perturbation analysis, which is an indispensable step in assessing the reliability of model explanations (Prümper et al., 2023). In the high-stakes context of education, reliability is paramount: explanations that vibrate significantly in response to minor input changes cannot be deemed trustworthy for instruction or intervention (Oluwatobi, 2025). The systematic perturbation analysis was applied to the XGBoost model using both SHAP and LIME, with the highly stable Logistic Regression results provided in **Appendix A.4** serving as a linear benchmark.

5.1.5.1 SHAP Explanation Analysis

The result of the SHAP (Lundberg and Lee, 2017) proved high stability, emphasizing their strength. As shown in Table 4, three main features - Num_summed_assessments, AVG_Score, and Num_OF_Prev_atttermpes - have corrected their rankings. The biggest absolute change has occurred in AVG_Score (= 0.0830), but this variant is still modest compared to its original value. Above all, all characteristics can still be understood and logically combined, confirming the overall explanation in stable predictions with high value. These results confirm that the SHAP is a reliable tool to understand the decisive logic of XGBOOST, even under small data disorders.

Table 4: SHAP Explanation Stability Analysis (XGBoost)

Rank	Original Feature	Original SHAP Value	Perturbed Feature	Perturbed SHAP Value	Absolute Change
1	num_submitted_assessments	5.7534	num_submitted_assessments	5.7755	0.0221
2	avg_score	1.2613	avg_score	1.1783	0.0830
3	num_of_prev_attempts	0.8769	num_of_prev_attempts	0.8765	0.0004
4	imd_band_80_90%	0.1629	total_clicks	0.1855	0.0226
5	highest_education_Lower_Than_A_Level	0.1369	imd_band_80_90%	0.1629	0.0260

5.1.5.2 LIME Explanation Analysis

On the other hand, explanations of LIME (Ribeiro et al., 2016) show significant instability. Table 5 shows that the initial prediction factor, Total_Clicks (+0.1161), has completely disappeared from the top year after disturbance, replaced by Num_Sumed_asssments (−0.1195). In addition, absolute changes for the main characteristics (0.2356 and 0.1887) show significant fluctuations. These changes show that the locally approximated method is difficult to capture complex and nonlinear interactions in XGBOOST (Slack et al., 2019). Therefore, its outcome may not be reliable for sensitive interventions at the student level where stability is essential.

Table 5: LIME Explanation Stability Analysis (XGBoost)

Rank	Original Feature	Original Contribution	Perturbed Feature	Perturbed Contribution	Absolute Change
1	total_clicks ≤ 849.97	+0.1161	num_submitted_assessments ∈ (4.00, 6.73]	−0.1195	0.2356
2	avg_score ∈ (68.00, 72.77]	−0.1121	gender_M	−0.1169	0.0049
3	gender_M	−0.1109	avg_score ∈ (68.00, 72.77]	−0.0905	0.0204
4	num_submitted_assessments ∈ (4.00, 6.73]	−0.1012	studied_credits > 120.00	+0.0876	0.1887
5	highest_education_Lower_Than_A_Level = 1	+0.0489	highest_education_Lower_Than_A_Level = 1	+0.0585	0.0096

The comparison emphasizes an important consideration for educational applications with high issues: SHAP provides strong and coherent explanations, stemming from a solid theoretical basis (Lundberg and Lee, 2017). LIME, on the contrary, has significant local instability, which makes its explanations unreliable to guide sensitive interventions (Salih et al., 2023). Data interference for logistics regression model, showing that the inherent stability is higher due to its linear nature, included in the Appendix for complete reference and confirmation of this conclusion.

5.1.6 Explanation Clustering and Student Personas (RQ 6)

This section interprets the Explanation Clustering findings, demonstrating how XAI outputs can be translated into actionable frameworks for targeted educational support.The explanations at the individual level of the XGBOOST model have been converted using the K-means cluster that is applied to SHAP values, providing a bridge to explain the technique of the model for solutions focusing on policies (Cooper, 2022).

As illustrated in Figure 12, the cluster algorithm has divided the student's body in three separate characters according to the similarity of the contribution of the characteristics. Checking the average formal values in each cluster allows a clear characteristic:

- Cluster 0 (Students focus on evaluation) mainly affected by learning performance, with num_submitted_assessments (average shap = 2.69) and AVG_Score (with the same meaning = 1.26) as predictive factors. For this group, the risk of dropping out is mainly determined by underestimating or poor performance, showing the necessity of the targeted learning interventions.
- Cluster 1 (students are at high risk and multiple factors) presents the highest overall Shap values, with num_submitted_assessments (mean SHAP = 5.58) and AVG_Score (the mean SHAP = 1.02) being the central feature. The combination of many academic factors and unfavorable behaviors shows that this cluster represents students with the highest risk, requiring in-depth support and many aspects.
- Clusters 2 (behavioral and demographic students) show more uniform dispersion functions. Academic data is still relevant, but behavioral activity (Total_Clicks = 0.15) and demographic characteristics (higher_eduction_lower_than_a_level = 0.18) plays a more important role, showing that interventions focus on participation or pastoralization may be the most effective.

In general, the cluster confirms the use of official values to divide students into categories that can be used, highlighting the uniformity of giving up and allowing educators to design targeted interventions based on evidence in accordance with the most influential forecast elements for each student type (Zarei and Pahl, 2025).

5.2 Comparison to Literature

5.2.1 Confirmations

The predictable efficiency of the models used in this study confirms the main trends in the project on predicting students' dropout. Suitable for Fernandes et al. (2020), the results show that automatic learning methods are effective in providing students' results. Specifically, the higher performance of XGBOOST compared to the logistics regression confirms the widespread observation that nonlinear general methods often surpass the simpler linear methods (Yang, 2021; Lopez et al., 2024). This is also consistent with results through studies on application education data exploration (Waheed et al., 2022),

This confirmation provides a solid technical basis but also highlights the limits of the study that focuses on performance. A large part of my dissertation emphasizes the prediction accuracy while ignoring the explanation. This thesis is consolidating that if a large accuracy can be achieved, deeper information about why the prediction is given is essential to convert the results into exploited educational strategies (Nagy and Molontay, 2023).

5.2.2 Contradictions and Novelty

The most significant contribution of this project lies in the dispute about the static view of the previous work. Studies like Wang et al. (2024) has developed strong time prediction models but often failed to interpret. By applying Shap(Lundberg and Lee, 2017) and LIME (Ribeiro et al)

The results showed a dynamic change in my project at the beginning of the semester, the demographic variables and the most influential, while later in the semester, the academic performance indicators, such as VLE's evaluation and commitment results, dominated. This dynamic explanation contrasts with the static analysis of existing documents, capable of simplify the process of abandoning students too much. The new insight here is that effective interventions not only depend on important features, but also on when they count the most in a student's learning path.

5.2.3 XAI Framework Novelty

This dissertation also contributes by advancing the methodological the use of Xai method in education. While foundational work by Ribeiro et al. (2016) and Lundberg & Lee (2017) introduced LIME and SHAP, most subsequent studies apply these methods in isolation and with limited scope (Salih et al., 2023). In contrast, this research integrates both methods into a comprehensive, multi-layered XAI framework .

More specifically, the SHAPand LIME have been used to support the three additional analysis:

(1) Audit equity and bias for sensitive characteristics (K. V. S. et al., 2025),
(2) robustness testing through perturbation analysis (Prümper et al., 2023; Slack et al., 2019), and
(3) clustering of SHAP values to derive interpretable student personas (Cooper, 2022). This goes beyond prior work, addressing gaps identified in the literature, such such as the lack of robust explanation validation (Oluwatobi, 2025), the absence of systematic fairness assessments (K. V. S. et al., 2025), and the difficulty of translating insights into actionable interventions (Zarei and Pahl, 2025; Cooper, 2022).

By combining the prediction of performance with a rich and multi-aspect XAI framework, my project shows a path towards ethical, transparent, and exploitable in education (paper position, 2025).

5.3 Implications for Practice

This section reflects the analysis results of real-world recommendations that can be used for educators and organizations. By taking advantage of the model's predictions, interpretations, temporary analysis, and groups of students, the following framework supports interventions focusing on data and people in higher education (Nagy and Molontay, 2023).

5.3.1 Timely Interventions

The analysis of time indicates that the factors are systematically affected on the implementation of reduced risks during the semester, confirming a dynamic approach to support (Wang et al., 2024). In the first weeks (1 -4), predictive characteristics are dominated by initial participation indicators and weekly. This requires early interventions to prioritize students' commitments, focus on promoting online participation, timely access to equipment and encouraging initial reviews. In contrast, the semester predictions at the end (week 5 to 26) are strongly dominated by functions related to evaluation (number_summated_assessments and AVG_Score). Therefore, the next support must focus on the school's performance, including personalized comments, guidance on goals and resources to improve the evaluation results. By arranging support time with dynamic evolutionary predictions, organizations can effectively optimize resources and allocate resources (Nagy and Molontay, 2023).

5.3.2 Targeted Support By Clustering Personas

The fact that the group of official values (Lundberg and Lee, 2017) has successfully identified separate student characters, allowing the necessary segment for sewing intervention strategies (Cooper, 2022):

- Cluster 0 (Assessment-Driven Students): Risk is primarily determined by academic performance. Interventions should focus on assessment completion support, structured study plans, and exam preparation workshops.
- . Cluster 1 (students are at high risk and multiple factors): This group shows the highest risk, depending on both commitments and learning measures. They require intensive support for many aspects, combining assessments with individual advisers and motivating training.
- Cluster 2 (behavioral and demographic students): Risks are associated with the blend of educational training and previous commitment. Interventions may include active awareness, course advice and support for previous knowledge implementation.

By targeting interventions to the specific needs of each persona, educators maximize the impact of support measures while minimizing unnecessary resource allocation (Zarei and Pahl, 2025).

5.4 Strengths and Limitations

A critical evaluation of the study highlights several methodological strengths as well as limitations that must be acknowledged to contextualize the findings.

5.4.1 Strengths of the Study

One principal strength lies in the comparative, multi-model design. The evaluation of logistics regression, XGBOOST, random forest, and MLP allows the assessment of the balance of interpretation and predictive performance (Lopez et al., 2024). Logistic regression provides a transparent reference facility, while XGBOOST gives high prediction accuracy, ensuring that the following XAI analyses are both strict and practical (Yang, 2021). Another noteworthy force is the development of the complete Xai frame.

By applying a SHAP (Lundberg and Lee, 2017) and LIME ((Ribeiro et al., 2016) Another contribution is the integration of temporary analysis and strength, introducing newness. The importance of monitoring 26 a week shows the dynamic nature of predictions such as demographics, evaluation, and commitment measures (Wang et al., 2024).

Perturbation analysis provides a systematic assessment of the stability of the explanations, confirming the strength of the SHAP while exposing the volatility of LIME (Oluwatobi, 2025). This double analysis consolidates the contribution to the field of educational data science that can be explained.

5.4.2 Limitations and Challenges

Despite these strengths, several limitations must be recognised. The application of the SHAP and LIME to the random forest model is calculated due to memory constraints and prolonged implementation time, illustrating that not all models are also compatible with existing XAI techniques (Salih et al., 2023).

The instability of LIME also represents a methodological limitation. Minor perturbations, such as a 5% increase in total_clicks, produced disproportionate changes in feature rankings (Slack et al., 2019), while temporal explanations became unreliable beyond the early weeks of analysis.

Finally, the question of generalizing data sets is still important. The results are exclusively taken from the Oulad data set (Kuzilek et al., 2017), reflecting the remote learning context. Therefore, it is necessary to be cautious in the process of external results of the traditional university environment (Waheed et al., 2022).

Chapter 6: Evaluation, reflection and conclusions

This final chapter summarizes the results of the project, reflects the achievements of the research, highlighting its contributions to Explainable Artificial Intelligence (XAI) in education, evaluating and criticizing the research process and describing guidelines for future work. I pointed out how the project solved questions and research objectives given in the Chapter 1, while recognising both strengths and limitations that inform the next stages of research and practical application in educational contexts.

6.1 Summary of achievements

This project has successfully developed a prediction framework for dropout students, integrated into a complete Xai method. Four machine learning models (logistics regression, random forest, multi -layer piercetron (MLP) and XGBOOST) have been systematically implemented and evaluated. Xgboost obtained the showing that performance (Roc-Auc = 0.9628), confirming its ability as a powerful prediction tool (Yang, 2021), while logistics regression (ROC-AC = 0.8683) provides a transparent basic facility for explanation and comparison. The evaluation of justification for the selection of these two models for deeper analysis, accuracy balance, and prediction explanation. A remarkable perception is the temporary analysis, proving that the characteristics of dynamic importance in the semester (Wang et al., 2024). Early predictions are based on demographic characteristics and previous learning measures, while the next predictions are increasingly affected by learning performance indicators, such as evaluation points and VLE engagement. This dynamic point of view progresses beyond the typical static analysis of the previous study, providing information that can be exploited for student behavior (Nagy and Molontay, 2023). Auditing the Fairness using the SHAP (Lundberg and Lee, 2017) and LIME(Ribeiro et al V. S. et al., 2025), highlighting the need to assess the systematic bias evaluation. Perturbation analysis confirmed the stability and reliability of SHAP explanations for XGBoost, whereas LIME demonstrated instability under minor input changes (Slack et al., 2019), indicating limitations in high-stakes decision-making contexts (Prümper et al., 2023).

Finally, the explanations for the SHAP of clusters allow us to identify different student characters on the basis of the risk of droupout records (Cooper, 2022). This new approach has filled the gap between technical results and practical interventions, allowing educators to target more effective support strategies according to specific risk factors (Zarei and Pahl, 2025).

6.2 Key Contributions

This research makes some important contributions. First, it sets a dynamic frame with importance over time, giving a practical representation of student trajectories. Second, it offers a comparative review of form and LIME on black box models (XGBOOST) and white models (logistics regression), showing the strength of SHAPand showing the limits of LIME(Salih et al., 2023). Thirdly, by integrating Fairness audit, robustness testing, and clusters, research promotes Ethical applications, management and practical applications of AI in education (Position Paper, 2025). Finally, the translation explained to the student characters providing a practical tool for educators, allowing interventions to be targeted based on interpratation.

6.3 Critical Self-Evaluation

The project has enhanced the importance of balancing the accuracy of prediction with the ability to interpret, fairness, and reliability. Challenges include calculation limits for random forest models and MLP (Swamy et al., 2022) and the instability of LIME explanations (Ribeiro et al., 2016). In the case of rehearsals, a greater emphasis will be placed on evolutionary methods and more effective time analysis methods to improve practical implementation.

6.4 Future work

Future research should focus on developing actual interventions for educators, taking advantage of temporary explanations to provide information that can be used to mitigate the risk of giving up. In addition, the method must be confirmed on different educational data sets to evaluate the overall . The expansion of the equity audit to evaluate the strategies of refining and the clusters of robust explanations in the powerful models will further improve the practical and ethical application of Xai in education, supporting targeted interpretation can be based on evidence in different learning contexts.

Conclusion

This dissertation shows that the predictive model, when integrated into the complete Xai frame, can provide accurate information, can be understood, andis morally responsible for the risk of abandoning students. By combining time analysis, equity audit, strong testing, and clusters, it offers methods and practices, establishing a solid basis for future research and application interventions in educational data science.

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Appendix

Code and Data Repository Access

The full project repository, including the final OULAD dataset and all analytical notebooks, is available via the following shared Google Drive link:

https://drive.google.com/drive/folders/179b8xF870x9oTHR51hmYIxFHrvJue6GR?usp=drive_link

The codebase consists of five sequential Jupyter Notebooks, ensuring full traceability of the research process: 1. Data Preprocessing, 2. Baseline Model, 3. Hyperparameter Tuning, 4. SHAP and LIME Implementation, and 5. XAI Implementation (which contains the final comparative, robustness, and clustering analyses). The accompanying README.md provides full instructions for execution

A.1 Exploratory Data Analysis (EDA) Visualizations

This section presents the essential figures from the initial data discovery, providing a statistical context and justifying key feature engineering choices.

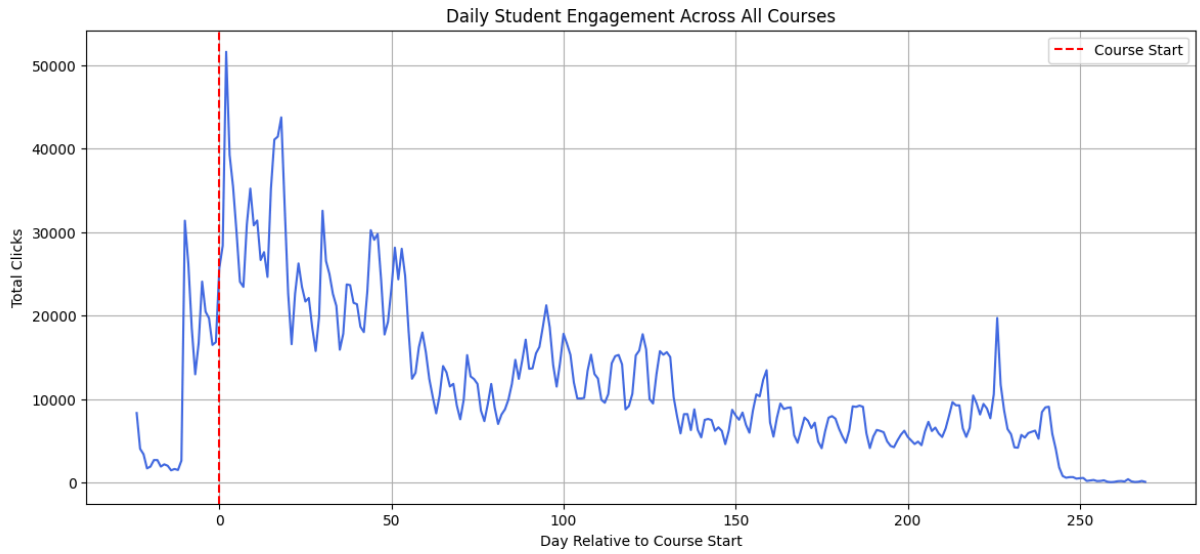

Figure A.1.1: A line plot showing the daily total VLE clicks across all courses, demonstrating the dynamic and non-uniform nature of student engagement over time.

Figure A.1 presents a line chart of the total daily VLE, illustrating the dynamic nature of the students' commitment. The X axis represents "the day compared to the start of the course", while the shaft displays "Total clicks". A red dot on day 0 marks the beginning of the official course. Commit to the peak right before and at first, then gradually decline. The weekly cycle peaks show the prediction research rhythm. This visualization emphasizes that students' commitments are more active than static. It directly justifies the techniques of the linked VLE data and sets the basic things of the temporary explanation analysis used later in dissertation.

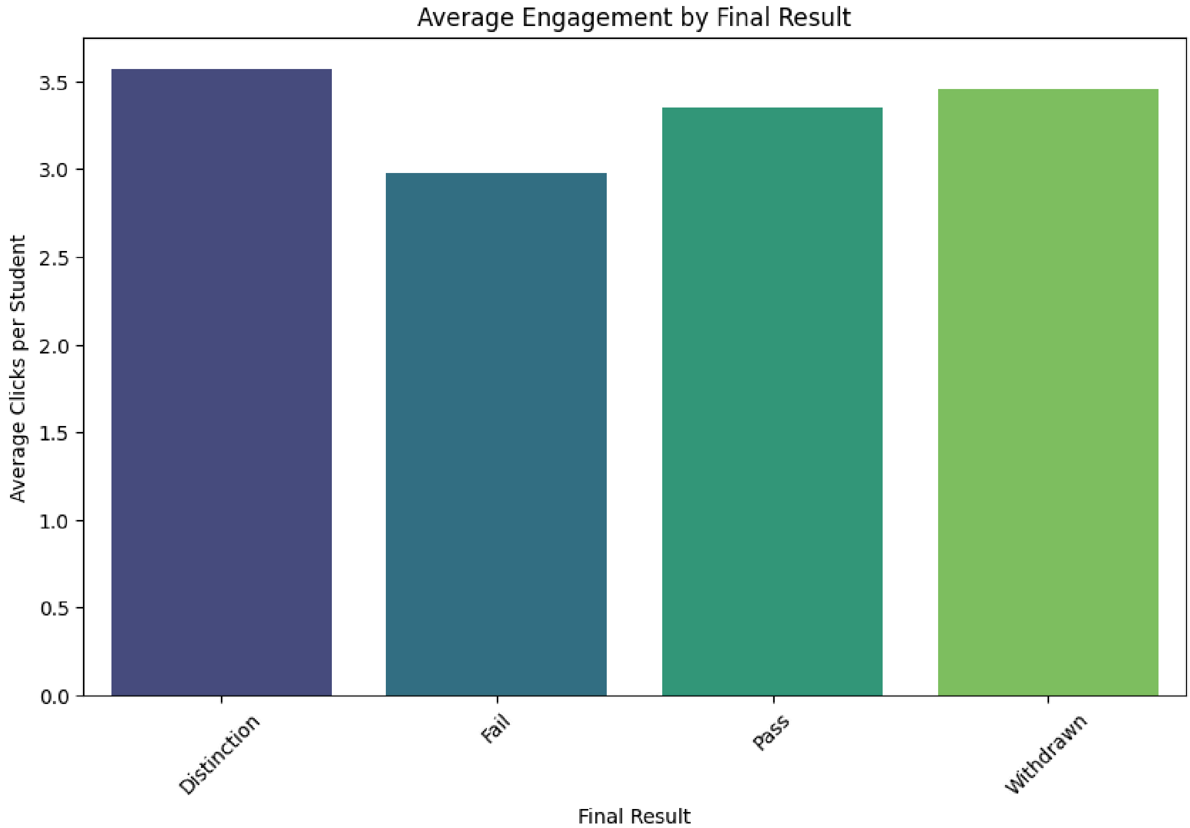

Figure A.1.2: A bar chart illustrating the relationship between average VLE clicks and the final course result, highlighting the complex relationship between engagement and success.

Bar graphics Figure A.2 Check VLE clicks between times compared to the final result of the course. The X-axis classifies the student according to Final_result (distinguish, overcome, fail, withdraw money) and displays the average clicks there. A higher commitment is observed for the difference, and students who pass are related to students who fail, but students who have withdrawn have an equally high commitment, which indicates that the commitment alone does not fully explain the results. This non-linear model highlights the complexity of students 'successful forecasts and justifies the use of advanced models and XAI techniques to explore nuances and unclear factors that stimulate students' performance beyond simple commitment data.

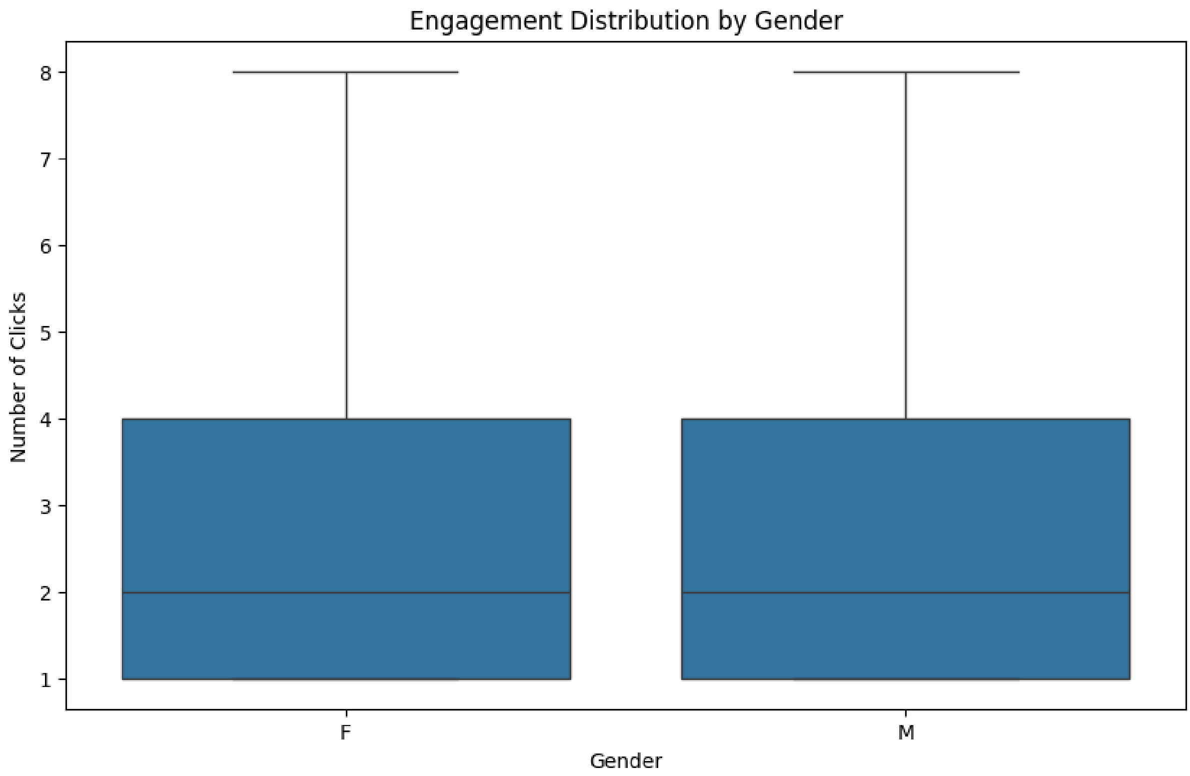

Figure A.1.3: A box plot showing the distribution of VLE clicks by gender, revealing no significant difference in engagement patterns between male and female students.

Box chart for comparison of VLE clicks between male (M) and female (F) students, Figure A.3. The Y-axis represents clicks, and the X-axis describes gender. The median, region, and total distribution were roughly identical, with no significant differences between genders. This insight provides a baseline of fairness and bias, but it does not show open gender differences in VLE activity. However, subsequent XAI analysis is necessary to determine whether the model can learn subtle gender-specific distortions through correlation properties, highlighting the importance of examining predictive behavior with regard to basic descriptive statistics.

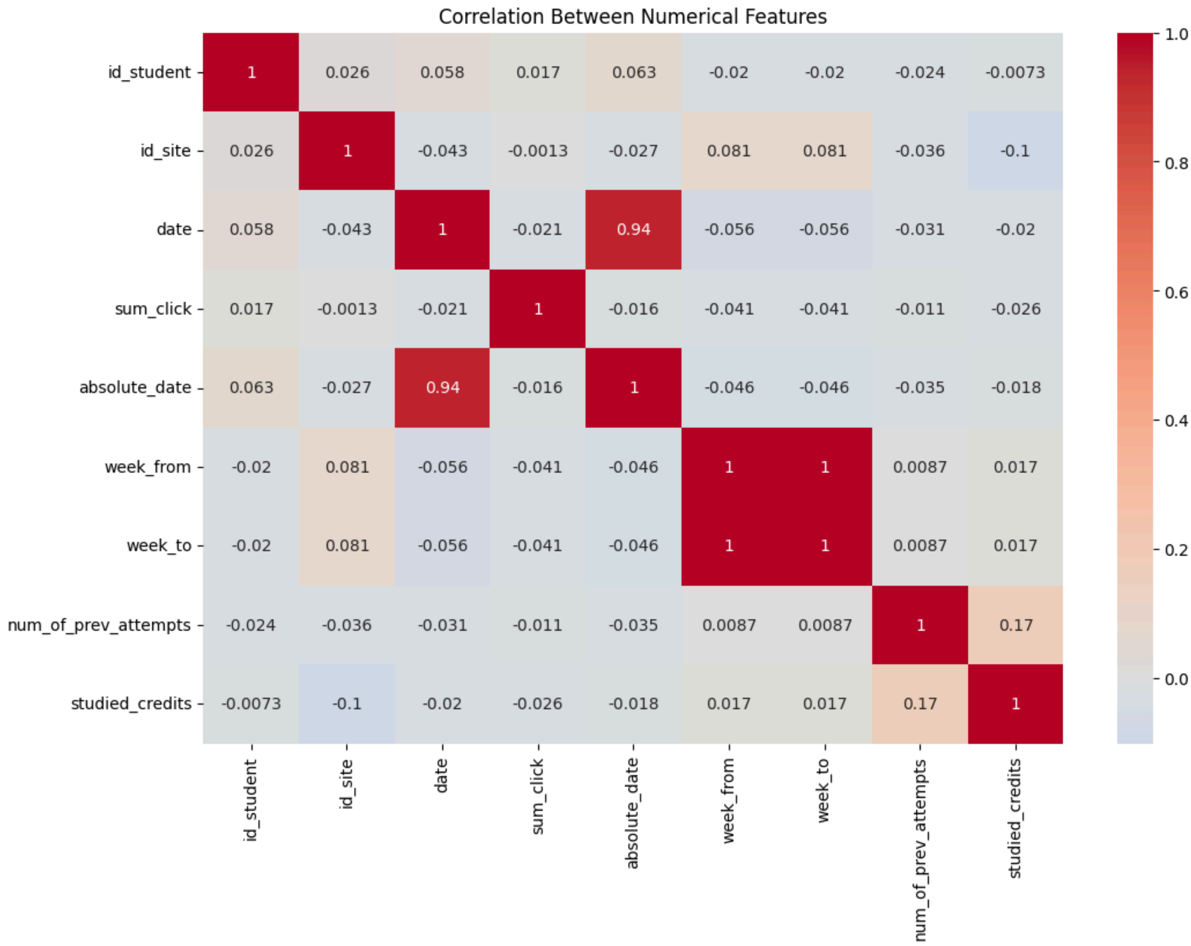

Figure A.1.4: A heatmap illustrating the linear correlation between key numerical features in the dataset, providing a holistic overview of data relationships.

The display of the heatmap in Figure A.4 of the linear correlation between numerical properties and numerical properties with color gradients from red to blue (strong negative). There is a strong correlation between date and Absolute_Date (0.94), and there is a perfect correlation between Week_from and Week_to (1.0), indicating redundancy. Slightly correlated features, such as Sum_Click, can have high model weights that show unique predictive signals. This visualization provides an overall view of the data record structure, provides information selection, and explains potential multicollinearity. In other words, the subsequent model-based characteristic meanings and XAI analysis are contextualized within the dissertation.

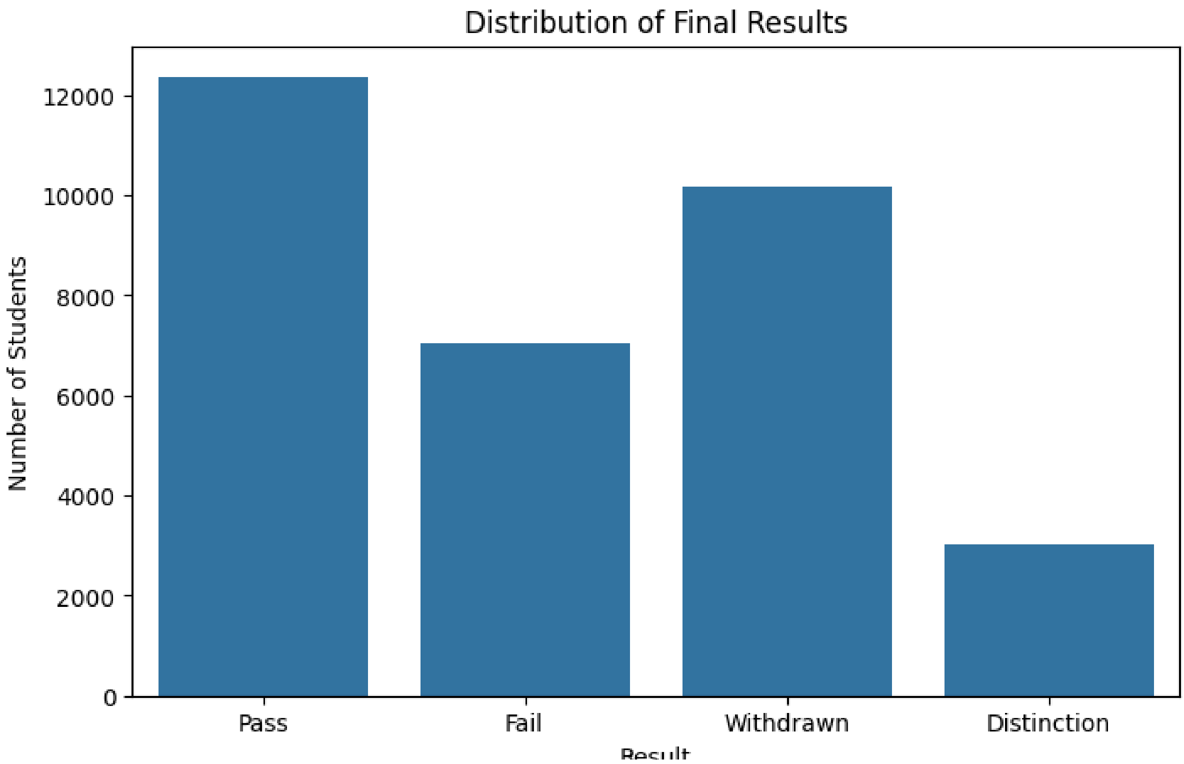

Figure A.1.5: A bar chart showing the distribution of the final student outcomes in the OULAD dataset over the number of students.

This bar chart, titled Distribution of Final Results is a crucial component of the Exploratory Data Analysis (EDA) findings. It serves to directly visualize the distribution of my target variable, final_result, which is a foundational step in understanding the problem you are trying to solve. The x-axis, labeled "Result," shows the four possible outcomes for students: Pass, Fail, Withdrawn, and Distinction. The y-axis, labeled Number of Students, indicates the count of students for each respective outcome. Figure A.5 reveals a significant imbalance in the dataset. While the number of students who passed is the highest (over 12,000), a substantial portion of students either failed or withdrew from the course. These two categories, which I have combined into my dropout target variable, represent a considerable portion of the data, but the class distribution is not uniform.

A.2 Temporal Explainability Outputs (RQ3)

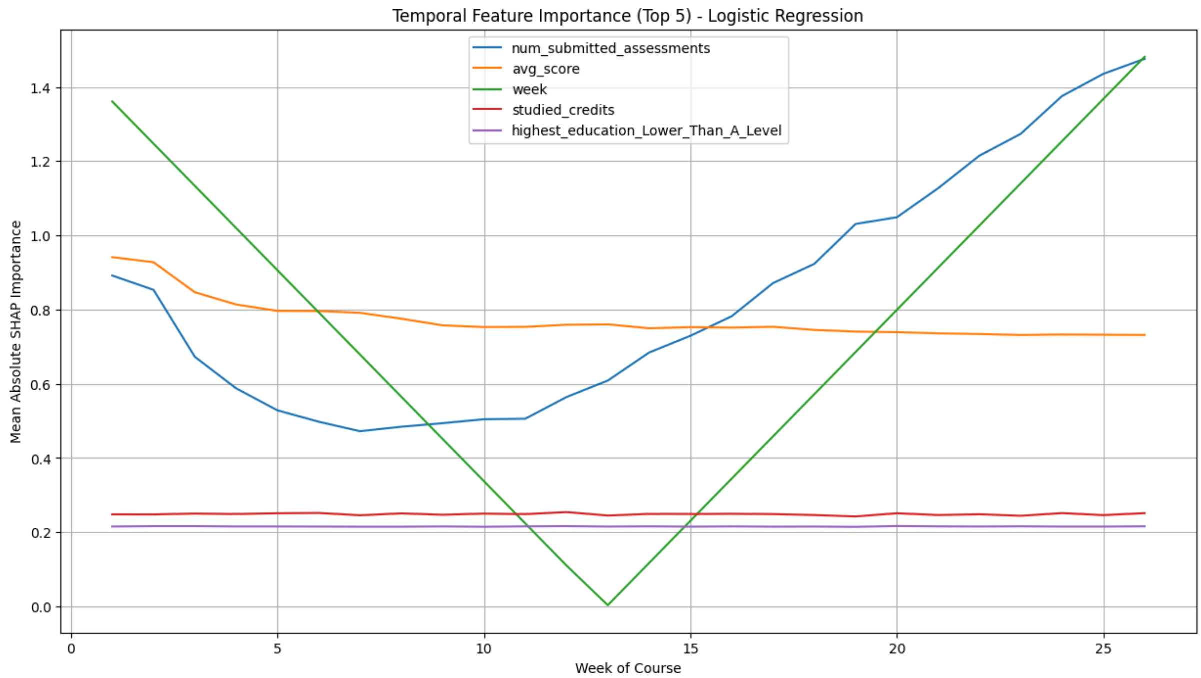

Figure A.2.1 is a line plot illustrating the temporal evolution of feature importance for the Logistic Regression model, as determined by the Mean Absolute SHAP Value.

The figure *A.2.1* displays the temporal function significance of the top 5 predictors for student dropout chance, as determined via way of means of SHAP values from a Logistic Regression model trained at the OULAD dataset. The x-axis represents the 26 weeks of the course, at the same time as the y-axis suggests the suggest absolute SHAP significance of every feature. The maximum influential capabilities are num_submitted_assessments, avg_score, and week, accompanied via way of means of studied_credits and highest_education (Lower Than A Level).

In deciphering those outcomes in terms of the study's question, we see that the predictive factors for the chance of scholar dropout aren't static, however, they evolve drastically throughout the semester. Early on, evaluation pastime and performance (num_submitted_assessments and avg_score) are key signals, suggesting that tracking those behaviors should permit for well-timed detection of at-chance college students withinside the first few weeks. We can see that the significance of num_submitted_assessments decreases in the early weeks, reaches its lowest point around week 7, and then progressively increases, becoming the dominant predictor by the later weeks. Average rating begins offevolved fantastically excessive however progressively declines in significance over time, at the same time as week suggests a V-shaped pattern, being much less crucial across the center of the semester however growing once more closer to the end. The different capabilities, studied credit and maximum training level, stay fantastically solid with low significance during the semester.

The locating that the predictive factors evolve means that educational interventions must be staged: early interventions must attention on evaluation engagement and scores, at the same time as later interventions should prioritize sustained participation and mitigating cumulative disengagement, reflecting that ongoing engagement is the most dependable predictor of persistence.

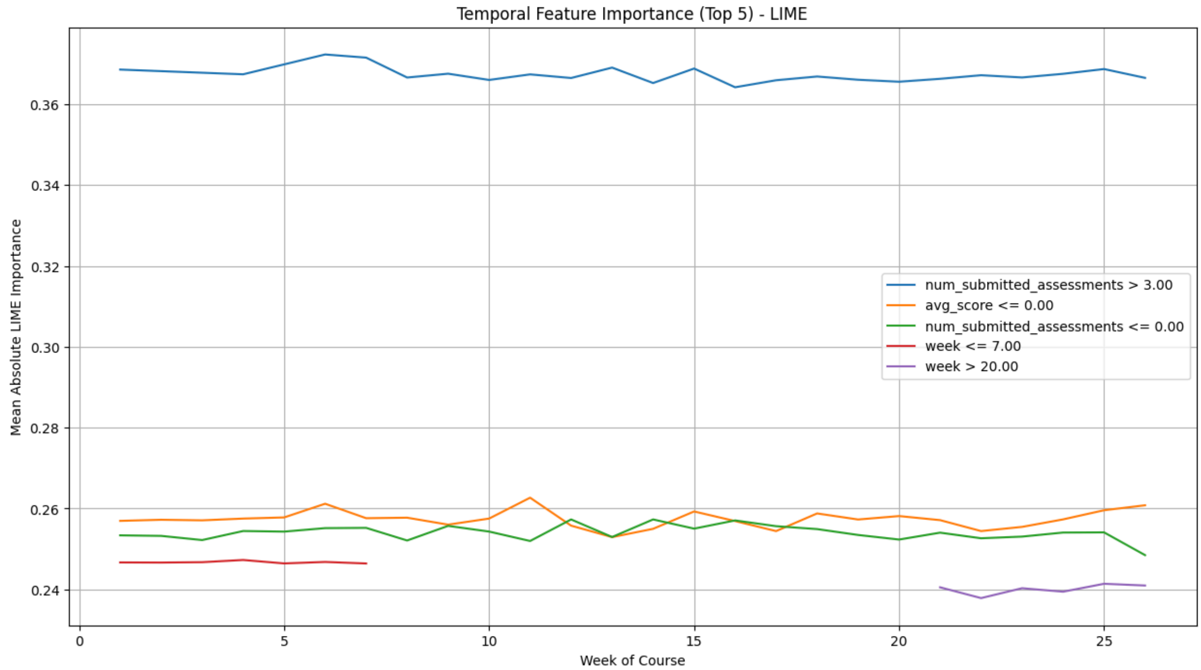

Figure A.2.2 is a line plot illustrating the temporal evolution of feature importance for the Logistic Regression model, as determined by the Mean Absolute LIME Value.

The figure A.2.2 i suggests the temporal function significance of the pinnacle 5 predictors for scholar dropout chance, derived from LIME factors implemented to a Logistic Regression version at the OULAD dataset. The x-axis represents the 26 weeks of the course, at the same time as the y-axis presentations the suggest absolute LIME significance of every function.

The key finding is that, like SHAP, which highlighted a dynamic evolution of function significance throughout time, LIME`s factors seem a ways greater solid. Among the capabilities, num_submitted_assessments > 3.00 genuinely dominates during the semester, with continually better significance than all others. The subsequent maximum influential predictors are avg_score

In interpreting these findings with respect to the research question, LIME's stability suggests that educators could reliably use assessment-related behaviors as steady indicators throughout the semester, without needing to heavily adjust focus across different time periods. However, this stability also reveals a **limitation**: LIME may underrepresent the dynamic temporal evolution of risk factors, thereby potentially overlooking critical moments when intervention timing could be most impactful. Thus, while LIME confirms the enduring importance of assessment engagement, SHAP's more dynamic temporal perspective (as seen in Figure*A.2.2)* provides additional insight into when interventions might be most effective.

A.3 Fairness and Bias Auditing Outputs (Research Question 4)

This section contains the detailed interpretations and raw data derived from the subgroup-specific LIME analyses for the XGBoost model. This data addresses the research question concerning potential algorithmic bias by examining how feature importance shifts across sensitive demographic categories.

```
  bst.update(dtrain, iteration=i, fobj=obj)
XGBoost model trained successfully.

Analyzing SHAP for subgroup: gender_M = 1.0...

Analyzing SHAP for subgroup: gender_M = 0.0...

Analyzing SHAP for subgroup: age_band_35_55 = 1.0...

Analyzing SHAP for subgroup: age_band_35_55 = 0.0...

Analyzing SHAP for subgroup: disability_Y = 0.0...

Analyzing SHAP for subgroup: disability_Y = 1.0...

--- Generating SHAP Plots and Values for Demographic Subgroups ---

Top 10 features for Subgroup: gender_1.0
                      Feature  Importance
0        num_of_prev_attempts     0.23769
1             studied_credits     0.23769
2                total_clicks     0.23769
3        num_vle_interactions     0.23769
4                   avg_score     0.23769
5   num_submitted_assessments     0.23769
6                    gender_M     0.23769
7  region_East_Midlands_Region    0.23769
8              region_Ireland     0.23769
9         region_London_Region    0.23769

Top 10 features for Subgroup: gender_0.0
                      Feature  Importance
0        num_of_prev_attempts     0.13192
1             studied_credits     0.13192
2                total_clicks     0.13192
3        num_vle_interactions     0.13192
4                   avg_score     0.13192
5   num_submitted_assessments     0.13192
6                    gender_M     0.13192
7  region_East_Midlands_Region    0.13192
8              region_Ireland     0.13192
9         region_London_Region    0.13192

Top 10 features for Subgroup: age_1.0
                      Feature  Importance
0        num_of_prev_attempts     0.23769
1             studied_credits     0.23769
2                total_clicks     0.23769
3        num_vle_interactions     0.23769
4                   avg_score     0.23769
5   num_submitted_assessments     0.23769
6                    gender_M     0.23769
7  region_East_Midlands_Region    0.23769
```

Figure A.3.1 is comparing the mean absolute LIME importance of the top 10 features across gender, age, and disability subgroups.

The figure (derived from the subgroup-particular LIME analyses) highlights the top 10 maximum essential features driving dropout predictions across sensitive demographic businesses, which include gender, age, and incapacity status.

In interpretation, those findings monitor how XAI strategies like SHAP and LIME can discover capacity algorithmic biases in predictive models. The consistency of instructional engagement capabilities throughout businesses indicates that the version`s primary selection drivers are behavior-based rather than demographic, which helps equity to an extent. The evaluation confirms that a wide variety of submitted assessments (each low,

≤four, and high, >9) constantly emerge because the most influential predictor, observed with the aid of common rating and overall clicks.

However, the repeated presence of subgroup identifiers consisting of gender, region, and age bands within the pinnacle causes shows that the version in part leverages touchy attributes. For instance, region-associated capabilities are greater in positive gender groups, at the same time as maximum education qualifications and age_band_55 seem variable among others. This indicates that at the same time as instructional engagement (submissions and performance) dominates prediction, demographic capabilities nevertheless play a measurable role in shaping version causes.

Ultimately, those subgroups suggest that equity auditing has to move past the combination function significance and study how causes shift throughout touchy classes to ensure equitable academic interventions. Using XAI in this manner allows researchers to audit models for bias, flag capabilities that can encode structural inequalities, and tell selections approximately whether or not to constrain or debias models before deployment.

A.3.1 Fairness and Bias Auditing Outputs LIME Features by Subgroup (XGBoost)

The LIME evaluation is well-known and shows the nearby reason of the high-appearing XGBoost version for unique pupil demographics (Gender, Age band, Disability status). This audit is essential for uncovering "diffused bias" that could be masked with the aid of using worldwide function significance metrics.

Table A.3.1: Top 10 LIME Features by Subgroup (XGBoost)

Subgroup	Rank	Feature	Importance
Gender (Male - 1.0)	1	num_submitted_assessments ≤4.00	0.556257
	2	num_submitted_assessments >9.00	0.466267

	3	avg_score ≤68.00	0.194626
	4	0.00<gender_M≤1.00	0.123959
Gender (Female - 0.0)	1	num_submitted_assessments ≤4.00	0.555002
	2	num_submitted_assessments >9.00	0.465740
	3	avg_score ≤68.00	0.190821
	4	gender_M ≤0.00	0.122943
Disability (No - 0.0)	1	num_submitted_assessments ≤4.00	0.558588
	2	num_submitted_assessments >9.00	0.465658
	3	avg_score ≤68.00	0.193313
	4	0.00<gender_M≤1.00	0.124551

The evaluation of the LIME outputs confirms that the version`s middle selection drivers are fair (assessment-based), but it is well-known that how sensitive features are applied locally:

- Core Fairness: Across all subgroups, the primary drivers are consistent: num_submitted_assessments (Importance ≈0.55) and num_submitted_assessments

> 9.00 (Importance ≈0.46). This is a fine finding, as the highest predictive weight is correctly tied to academic engagement, not demographics.

- Subgroup Influence: The touchy capabilities (gender_M conditions) constantly seem within the pinnacle four or five positions throughout all businesses, indicating a continuous, non-zero reliance on gender by the XGBoost model. For instance, the Gender (Male - 1.0) subgroup suggests that the particular gender situation has a significance of 0.123959, confirming that this sensitive attribute performs a measurable, nearby position within the prediction along instructional factors.
- Actionable Insight: The balance of the pinnacle drivers throughout subgroups indicates that interventions concentrated on low submission quotes can be widely powerful and fair, however, the routine presence of demographic capabilities necessitates continued tracking before deployment. This dual attention is fundamental to moral XAI.

A.3.2 Fairness and Bias Auditing Outputs Logistic Regression (RQ 4)

This segment consists of the distinct interpretations and raw statistics derived from the subgroup-specific LIME analyses for the Logistic Regression model. These data address the study's query regarding capability algorithmic bias through analyzing how characteristic significance shifts throughout touchy demographic categories.

The LIME evaluation found the nearby purpose of the Logistic Regression version for extraordinary scholar demographics. This audit is vital for uncovering "diffused bias" that might be masked through international characteristic significance metrics.

Note on Warnings

```
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
Top 10 LIME features for Subgroup: age_1.0
                                  Feature  Importance
0           num_submitted_assessments > 9.00    0.438752
27             num_vle_interactions > 253.46    0.392595
1             num_vle_interactions <= 253.46    0.381413
22         num_submitted_assessments <= 4.00    0.363652
23                        avg_score <= 68.00    0.263329
2                          avg_score > 82.33    0.226040
28                      total_clicks > 849.97    0.223841
3                      total_clicks <= 849.97    0.203062
18                   studied_credits > 120.00    0.146746
13  6.73 < num_submitted_assessments <= 9.00    0.116525

Analyzing LIME for Subgroup: age_0.0...
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
```

During the LIME implementation, the version regularly output warnings concerning lacking characteristic names (UserWarning: X does now no longer have legitimate characteristic names...). This is a recognized trouble while LIME converts the nearby perturbed statistics (which won't hold specific characteristic names) earlier than feeding it again into the scikit-learn version, which became outfitted with named functions. These warnings had been deemed non-vital to the correctness of the LIME contribution values; however, they spotlight the important statistics adjustments concerned in version-agnostic explainability.

Table A.3.2 : Top 10 LIME Features by Subgroup (Logistic Regression)

Subgroup	Rank	Feature	Importance
Age (Under 35)	1	num_submitted_assessments >9.00	0.438752
	2	num_vle_interactions >253.46	0.392595
	3	num_vle_interactions ≤253.46	0.381413
	4	num_submitted_assessments ≤4.00	0.363652
	5	avg_score ≤68.00	0.263329
	6	avg_score >82.33	0.226040
	7	total_clicks >849.97	0.223841
	8	total_clicks ≤849.97	0.203062

	9	studied_credits >120.00	0.146746
Age (35 and Over)	1	num_submitted_assessments >9.00	0.438752
	2	num_vle_interactions >253.46	0.392595
	3	num_vle_interactions ≤253.46	0.381413
	4	num_submitted_assessments ≤4.00	0.363652
	5	avg_score ≤68.00	0.263329
	6	avg_score >82.33	0.226040
	7	total_clicks >849.97	0.223841
	8	total_clicks ≤849.97	0.203062
	9	studied_credits >120.00	0.146746

The LIME evaluation on Logistic Regression is famous for its sturdy reliance on behavioral patterns. The Primarly functions are continuously associated with evaluation submission activity (num_submitted_assessments) and VLE engagement (num_vle_interactions, total_clicks). This validates the version as a behavior-based predictor. Because LIME`s method for linear functions is to become aware of those excessive/low thresholds, the version specializes in segments of the scholar populace described through those precise values (e.g., college students with low clicks or excessive submissions).

Crucially, the contrast between the 2 age subgroups indicates that the characteristic significance rating is sort of equal. This shows that, not like the XGBoost findings, the Logistic Regression version is based at the identical center set of functions to are of whether or not the scholar is below 35 or 35 and over. This tension confirms its position as a stable, interpretable benchmark, demonstrating that the linear version avoids the sort of diffused, localized bias that complicated fashions might also additionally study.

A.4 Robustness Analysis Outputs (RQ5)

The stability analysis for the Logistic Regression model provides critical context for evaluating the reliability of the XAI methodology. This analysis directly addresses the research question concerning the trustworthiness of explanations provided by a transparent model under data perturbation.

```
Logistic Regression model trained successfully.

Analyzing explanation for student at index 50...
Original Prediction: Dropout

--- Original SHAP Explanation (Top 5) ---
                                 Feature  SHAP_Value
0                   num_of_prev_attempts    0.953643
1                        studied_credits    0.515266
5                  num_submitted_assessments    0.353658
20  highest_education_Lower_Than_A_Level    0.323219
6                               gender_M    0.057351

Perturbed Prediction: Dropout

--- Perturbed SHAP Explanation (Top 5) ---
                                 Feature  SHAP_Value
0                   num_of_prev_attempts    0.953643
1                        studied_credits    0.515266
5                  num_submitted_assessments    0.353658
20  highest_education_Lower_Than_A_Level    0.323219
6                               gender_M    0.057351

--- Comparison of Explanations ---
Original top 5 features:
                                 Feature  SHAP_Value
0                   num_of_prev_attempts    0.953643
1                        studied_credits    0.515266
5                  num_submitted_assessments    0.353658
20  highest_education_Lower_Than_A_Level    0.323219
6                               gender_M    0.057351

Perturbed top 5 features:
                                 Feature  SHAP_Value
0                   num_of_prev_attempts    0.953643
1                        studied_credits    0.515266
5                  num_submitted_assessments    0.353658
20  highest_education_Lower_Than_A_Level    0.323219
6                               gender_M    0.057351
```

Figure A.4.1: SHAP Explanation Robustness Analysis for Logistic Regression, showing the perfect consistency of the top 5 features before and after a 5% perturbation of the `total_clicks` feature.

The picture output you provided (Figure B.3.1) illustrates the center of the perturbation evaluation. An unmarried, consultant scholar's document became selected, and a minimal, practical extra a 5% boost in the total_clicks characteristic was introduced. The novelty lies in checking out whether or not the reason from this linear version might extrapolate even if perturbed on a non-linear input characteristic.

Key Findings and Interpretation: The effects display an incredibly excessive degree of stability, as proven through the contrast of the Original and Perturbed SHAP Explanations within the figure.

- Perfect Ranking Consistency: The rating of the pinnacle 5 functions—num_of_prev_attempts, studied_credits, num_submitted_assessments, highest_education_Lower_Than_A_Level, and gender_M—remained equal after the perturbation.
- Minimal Magnitude Change: The SHAP values confirmed near-0 extrade in magnitude. For example, num_of_prev_attempts maintained its impact at about 0.9536, and studied_credits remained at 0.5152.

This locating is notably significant. It confirms that the Logistic Regression version's inner shape is so inflexible and linear that a minor extrade in a unmarried enter characteristic is inadequate to regulate its essential decision-making hierarchy. This reliability validates the use of Logistic Regression as the perfect "white-box" benchmark against which the stability and potentially chaotic behavior of the complex XGBoost model must be measured. This robust stability enhances the trustworthiness of the insights derived from this model for informing high-stakes educational decisions.

```
Analyzing explanation for student at index 50...
Original Prediction: Dropout
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(

--- Original LIME Explanation (Top 5) ---
                                         Feature  Contribution
0                   num_vle_interactions <= 253.46      0.395787
1                          total_clicks <= 849.97     -0.240742
2                        studied_credits > 120.00      0.142341
3           4.00 < num_submitted_assessments <= 6.73      0.094832
4  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.084028

Perturbed Prediction: Dropout
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LogisticRegression was fitted with feature names
  warnings.warn(

--- Perturbed LIME Explanation (Top 5) ---
                                         Feature  Contribution
0                   num_vle_interactions <= 253.46      0.374197
1                           total_clicks > 849.97      0.193897
2                        studied_credits > 120.00      0.142600
3  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.088296
4           4.00 < num_submitted_assessments <= 6.73      0.087893

--- Comparison of Explanations ---
Original LIME Explanation:
                                         Feature  Contribution
0                   num_vle_interactions <= 253.46      0.395787
1                          total_clicks <= 849.97     -0.240742
2                        studied_credits > 120.00      0.142341
3           4.00 < num_submitted_assessments <= 6.73      0.094832
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Perturbed LIME Explanation:
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0                   num_vle_interactions <= 253.46      0.374197
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3  0.00 < highest_education_Lower_Than_A_Level <= 1.00      0.088296
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```

Figure A.4.2: SHAP Explanation Robustness Analysis for Logistic Regression, showing the perfect consistency of the top 5 features before and after a 5% perturbation of the total_clicks feature.

Stability analysis of logistics regression models provides an important context for assessing the reliability of XAI methodology. This analysis directly addresses research questions regarding the reliability of explanations provided by transparent models under data failures. Methodology and novelty: The analysis shown in Figure A.4.2 focuses on creating a disturbed version of a representative student data record (index 50) by introducing the Total_Clicks function by 5%. The novelty of this approach is to empirically test whether explanations from linear interpretable models exhibit changes when hampered by nonlinear input functions that test endogenous stability.

The results show a very high level of stability and confirm the robust nature of the internal structure of the logistics regression model. Prediction consumption: The original prediction was a dropout, and the distorted prediction remained a dropout. This points out that model classification decisions are not affected by changes in input. Complete Ranking Consistency:

As seen in the illustration, the ranking of the 5 best features - num_of_prev_tempts, submitted_credits, num_submited_assessments, hegst_education_lower_than_a_level, and gender_m Minimum Size Changes: Shape Values indicate the size of extreme zero variations, ensuring that the weights of the model are stiff and stable. For example, num_of_prev_tempts maintains its effect to 0.953643 in both scenarios. This finding is extremely important. We confirm that the linear and rigid structure of the logistics regression model is strongly stable against minor data variations and therefore validate the advantage as a complete "white box" benchmark for the entire XaI comparison. This robust stability improves the reliability of the findings derived from this model to inform very old educational decisions.